UE23CS352B - Object Oriented Analysis & Design

Mini Project Report

Smart Campus Event Management System: A Role-Based Web Application for University Event Automation

Submitted by:Keshav Vithal Sangonda

Team Members

NAME	SRN
Keshav Vithal Sangonda	PES1UG23CS300
Pushkar P	PES1UG24CS822
Jeevan A N	PES1UG23CS266
Kasa Guruvi Reddy	PES1UG23CS289

6TH Semester - "E" Section

Prof.Vinay Joshi

January - May 2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act
No. 16 of 2013) 100ft Ring Road,
Bengaluru – 560 085, Karnataka, India

Problem Statement: Smart Campus Event Management System

Key Features:

1. Comprehensive Role-Based Access Control (RBAC):

- Implemented distinct dashboards and secure access levels for four primary actor roles: Student, Organizer, Faculty, and Admin.
- Each role is restricted to its specific workflows (e.g., Admins manage users/payments, Faculty review events, Organizers host events, Students register).

2. Complete Event Lifecycle Management:

- Creation & Proposal: Organizers can draft and propose new campus events with specific details, dates, and media.
- Faculty Approval System: Implemented a robust approval workflow where Faculty members must review, approve, or reject proposed events before they are visible to students.
- Date Restriction Logic: Implemented dynamic date validation to ensure events are created with logical, future timelines.
- Modification/Deletion: Organizers retain control to edit event details or delete events depending on their current approval status.

3. Advanced Smart Registration & Eligibility Filtering:

- Developed a smart registration engine that evaluates students' profiles against event prerequisites.
- Automatically filters out and denies registration to students who do not meet specific CGPA or Age limits set by the organizer.

4. Integrated Payment-to-Certificate Workflow:

- Built a streamlined process for paid events. The system tracks the financial transaction, updating the student's status upon successful payment, and links this directly to their eligibility for final event certification.

5. Dedicated Attendance Tracking System:

- Provided organizers with a dedicated interface to verify and mark the physical or virtual attendance of registered students on the day of the event.

6. Automated PDF Certificate Generation:

- Implemented a system that automatically designs and generates custom PDF certificates upon event completion.
- Strict validation ensures that certificates are generated only for students who were explicitly marked as "Attended" by the organizer.

7. Secure Authentication & Account Recovery:

- Implemented secure user login and session management.
- Integrated an automated Password Recovery Workflow via Gmail SMTP, allowing users to securely reset forgotten passwords via email verification.

8. Professional & Dynamic User Interface:

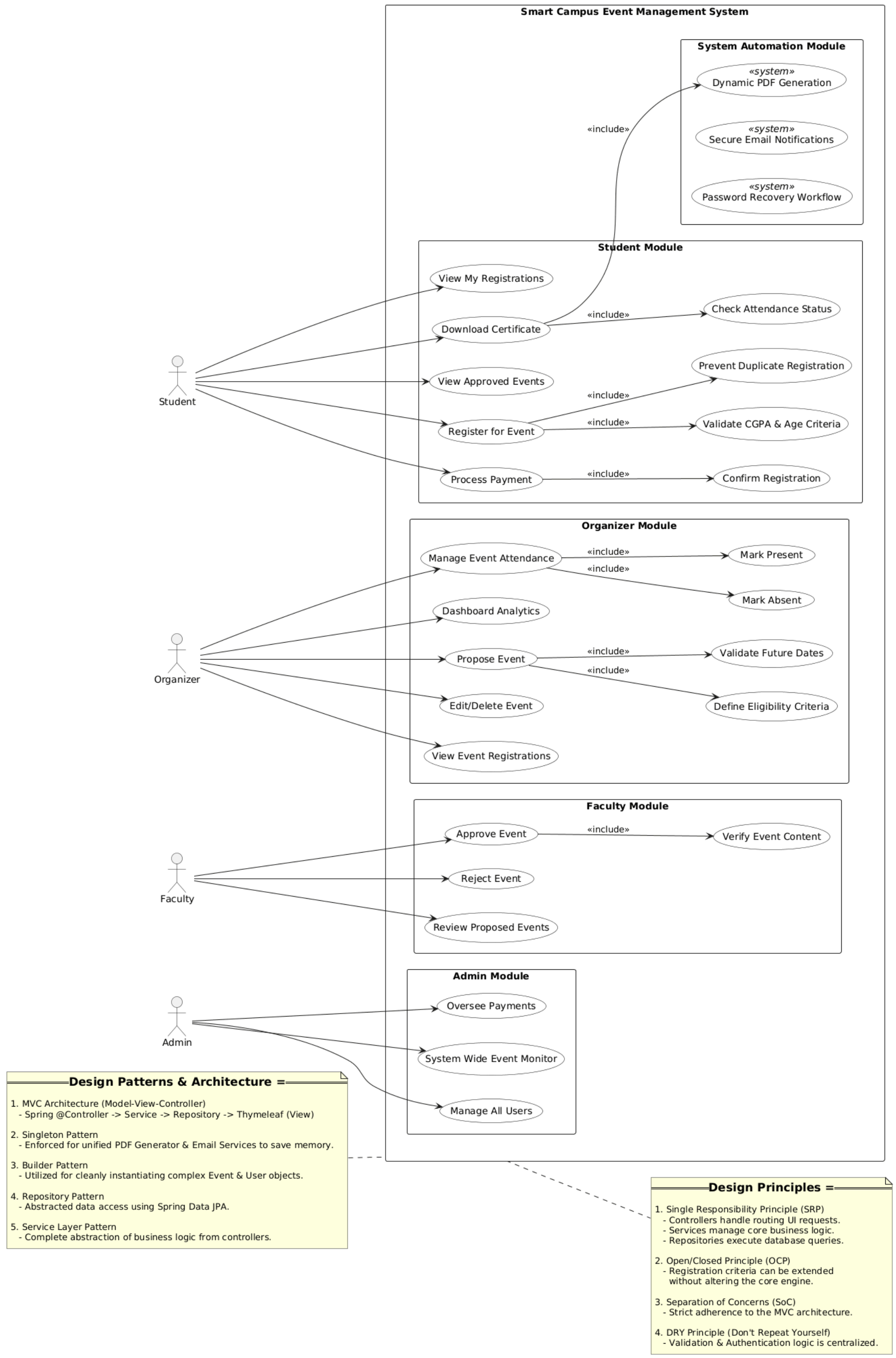
- Built a responsive, aesthetically pleasing frontend using modern web development standards.
- Included support for dynamic background styling, event image uploads, and visual flair to emulate a premium, real-world application.
- Included static informational architecture such as About Us and Privacy Policy pages to make the application legally and professionally complete.

9. Application of Software Design Patterns:

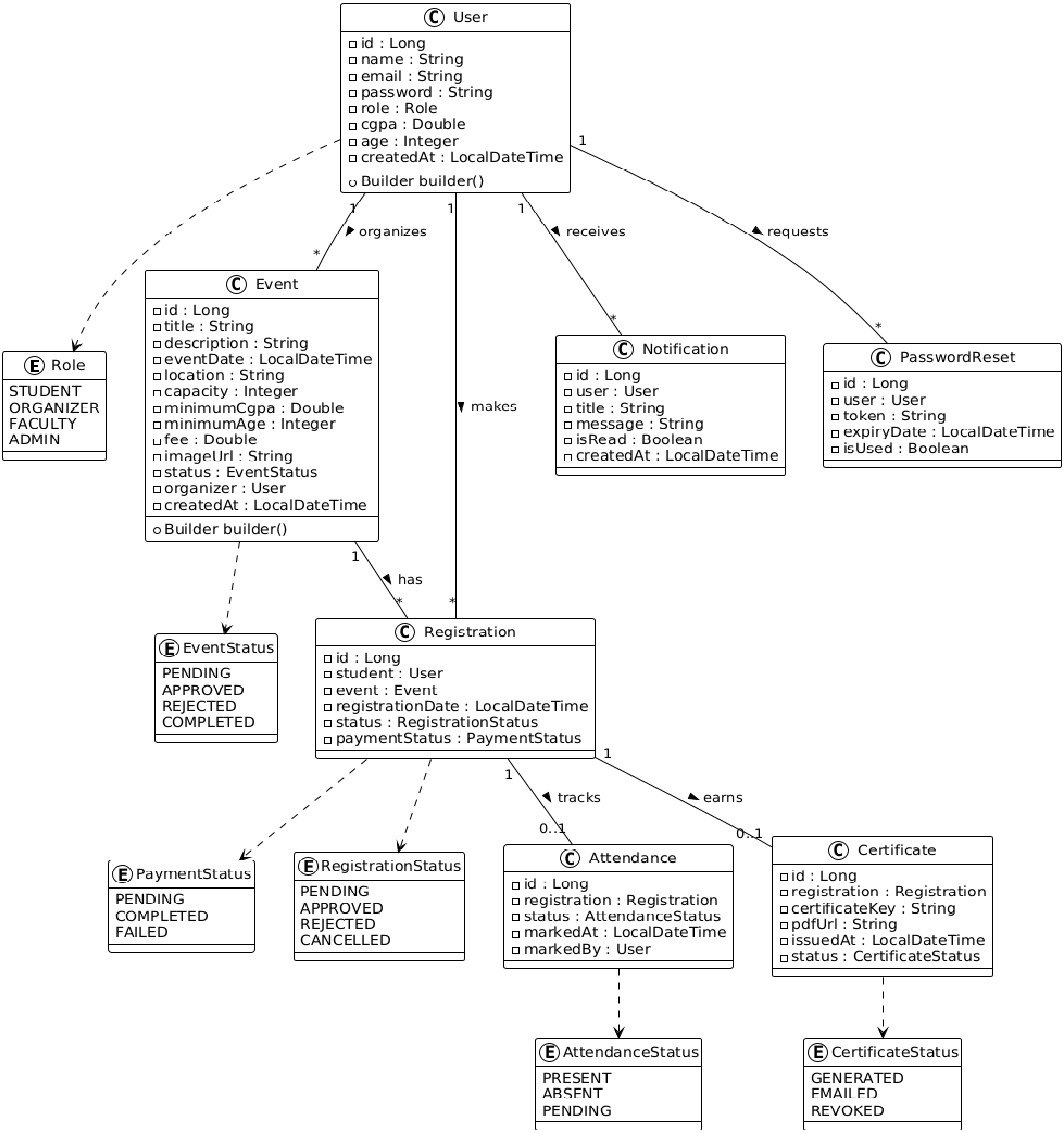
- Structured the backend utilizing industry-standard Object-Oriented Analysis and Design (OOAD) principles.
- Specifically implemented the Singleton Pattern and Builder Pattern to construct robust, scalable, and maintainable Java code.

Models:

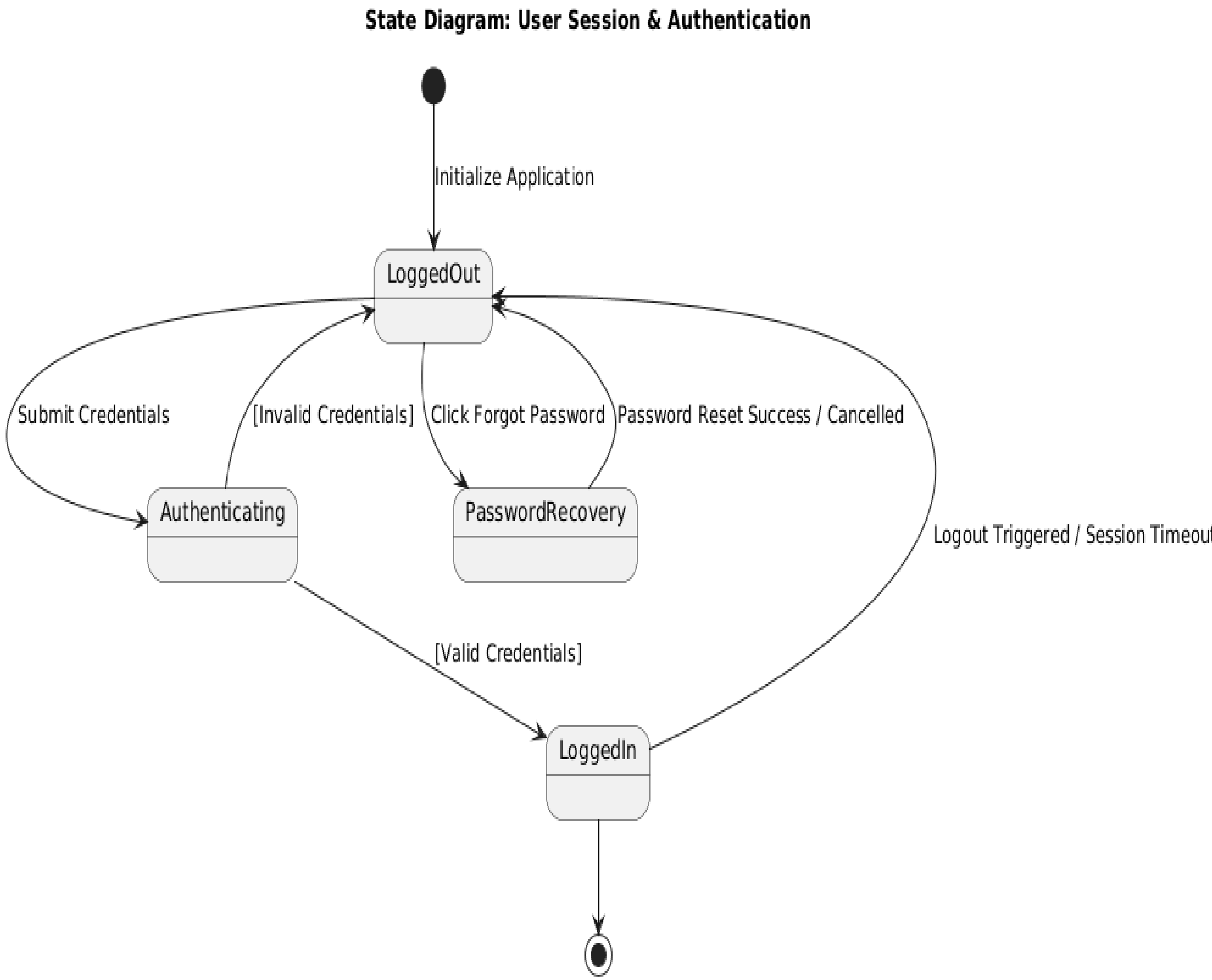
Use Case Diagram:



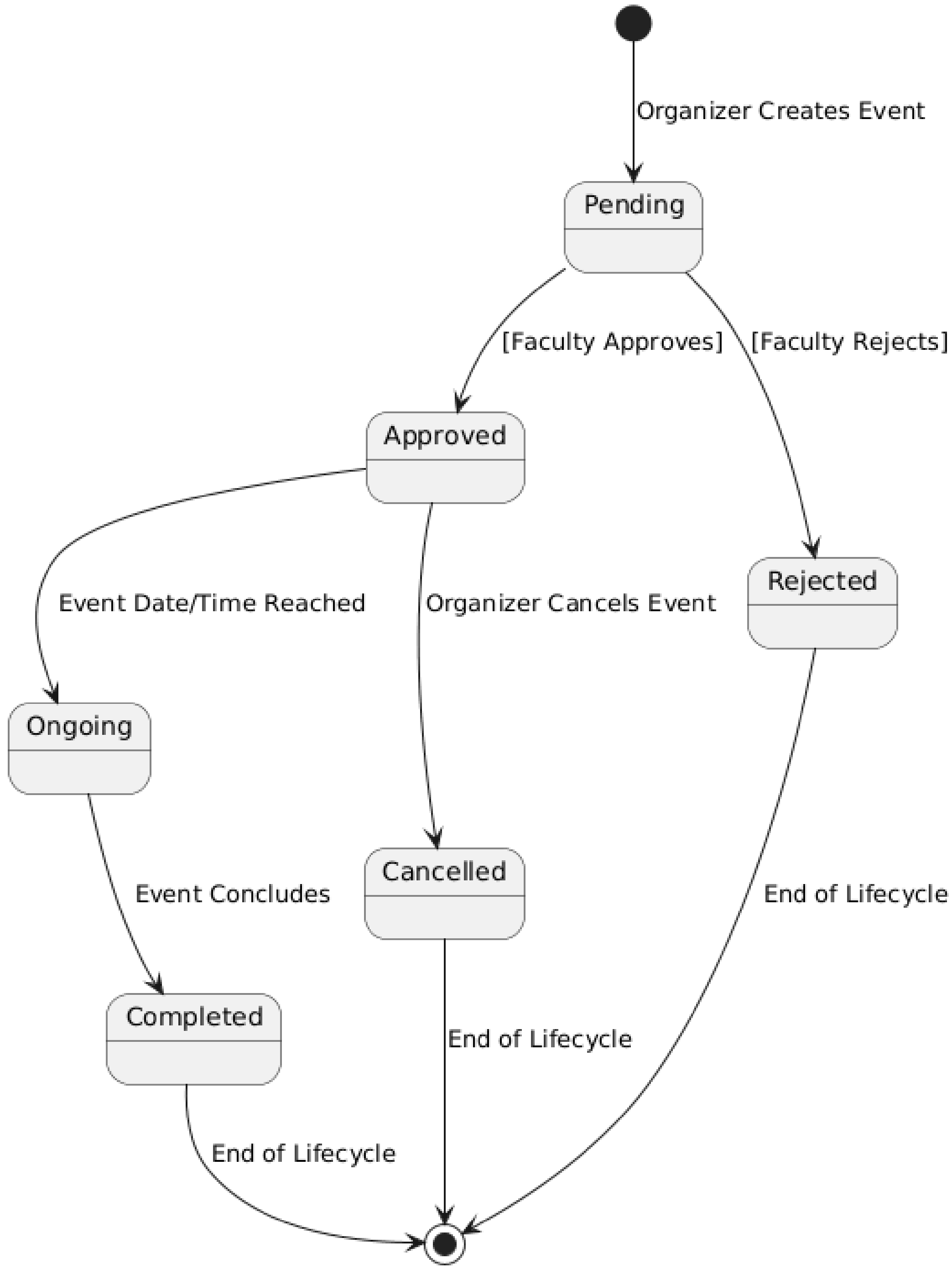
Class Diagram:



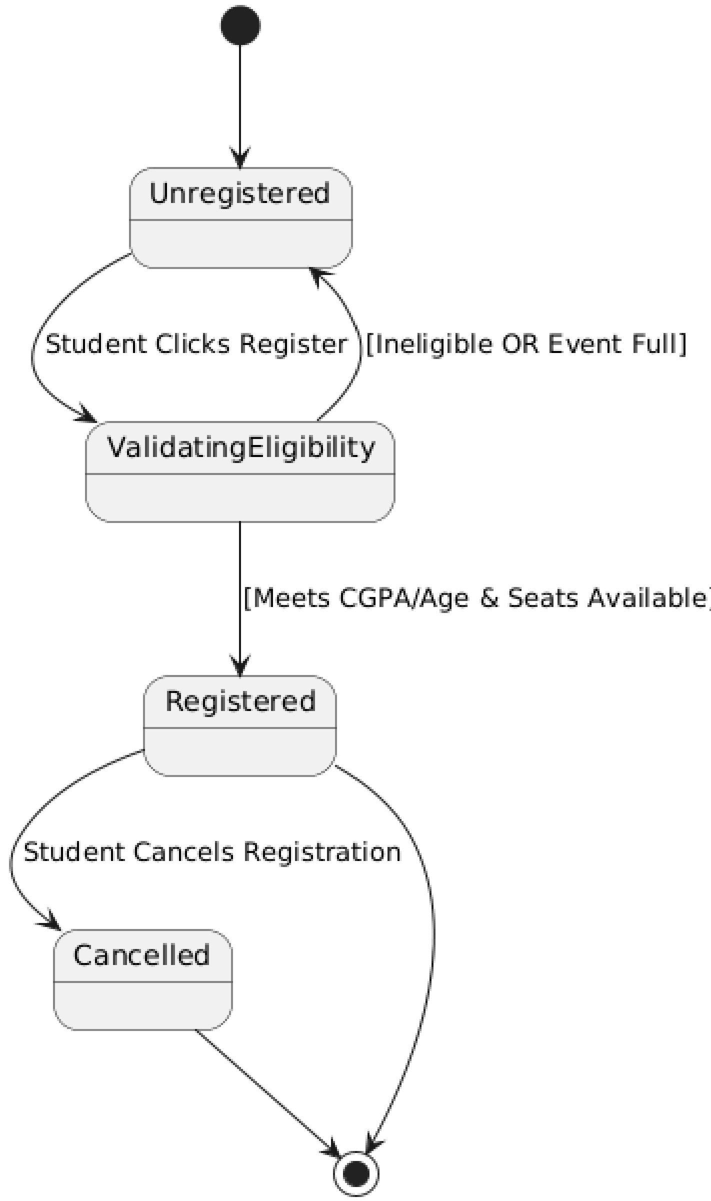
State Diagram:



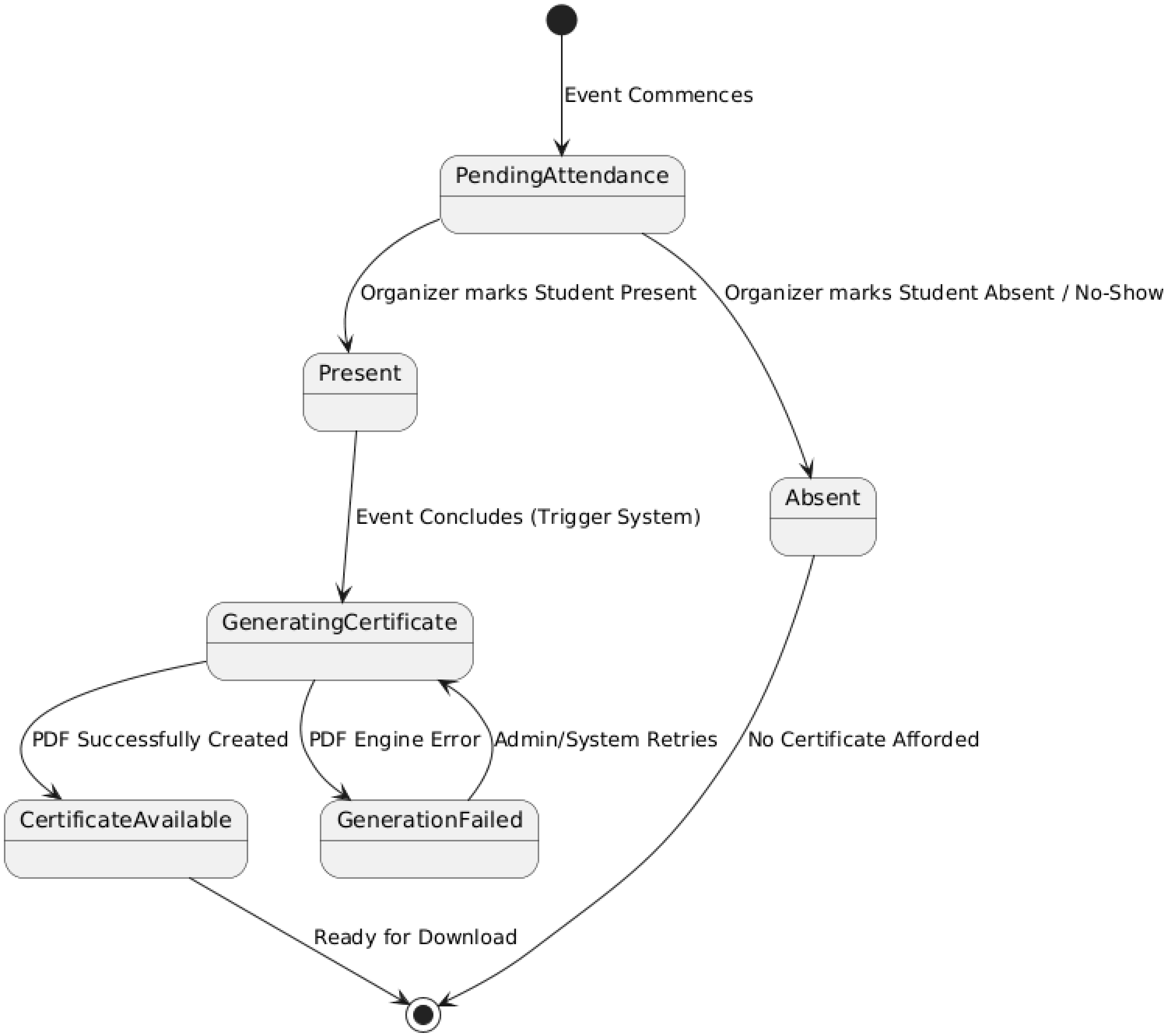
State Diagram: Event Lifecycle



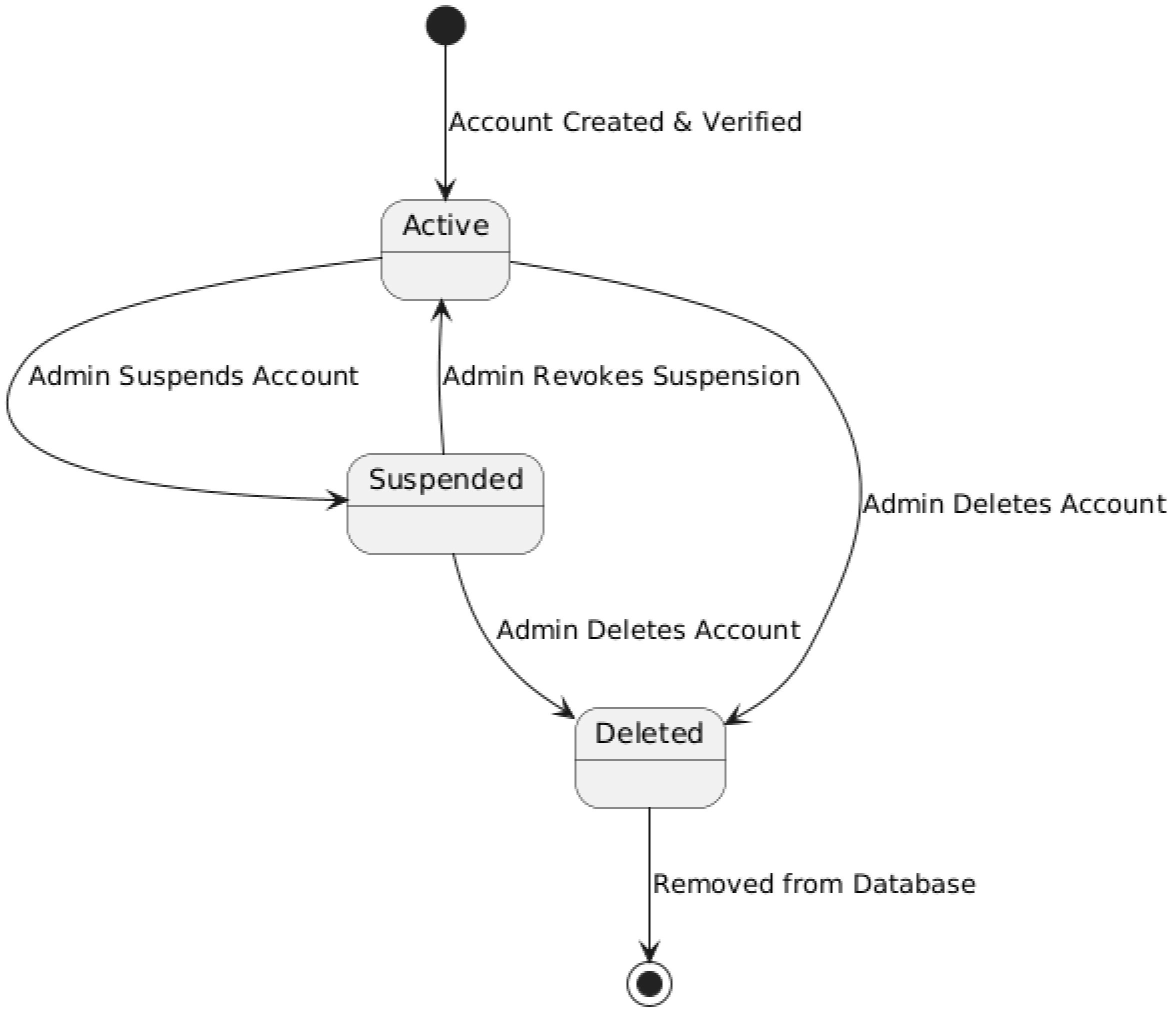
State Diagram: Student Registration Ticket



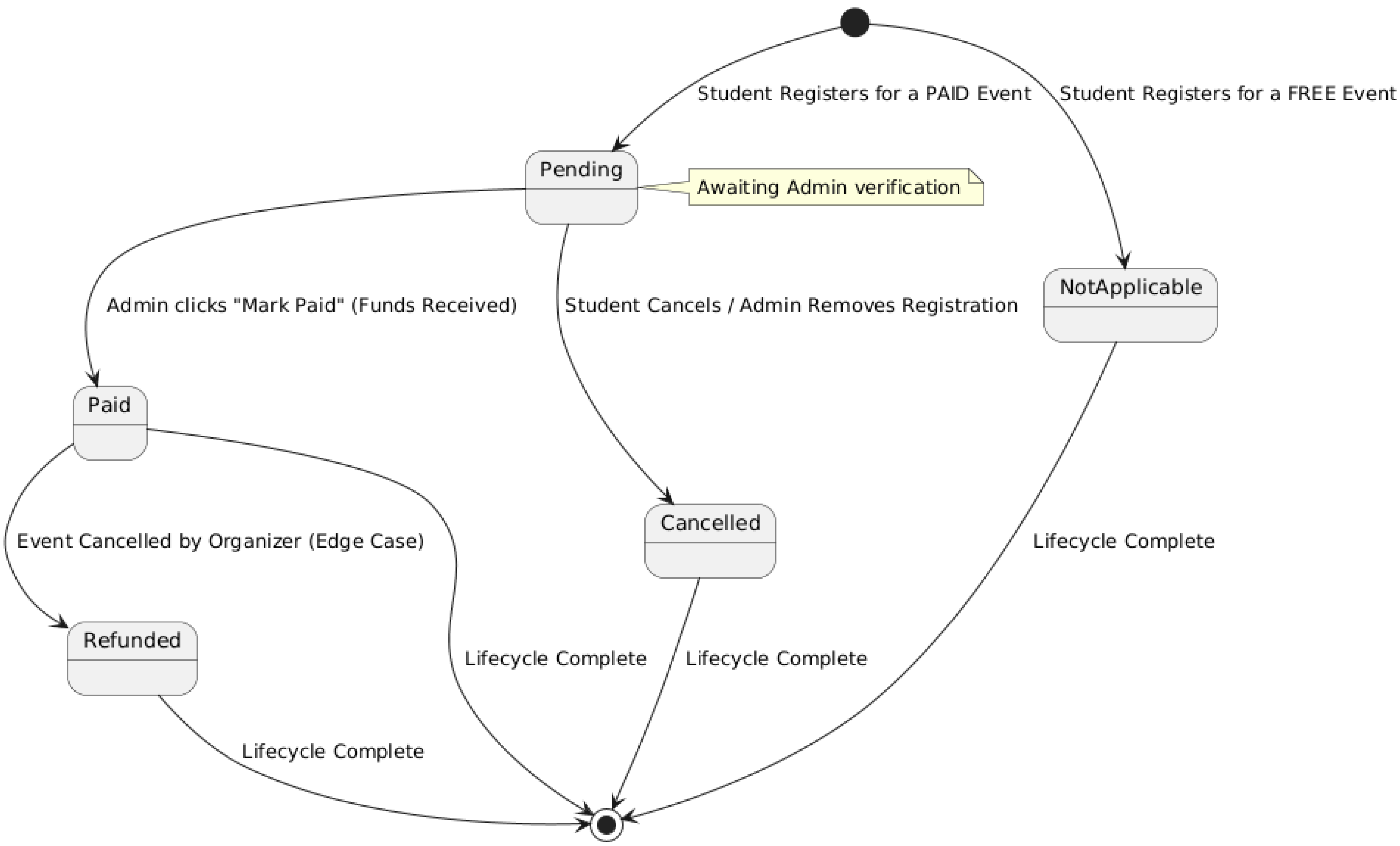
State Diagram: Attendance & Certificate Generation



State Diagram: User Account Status

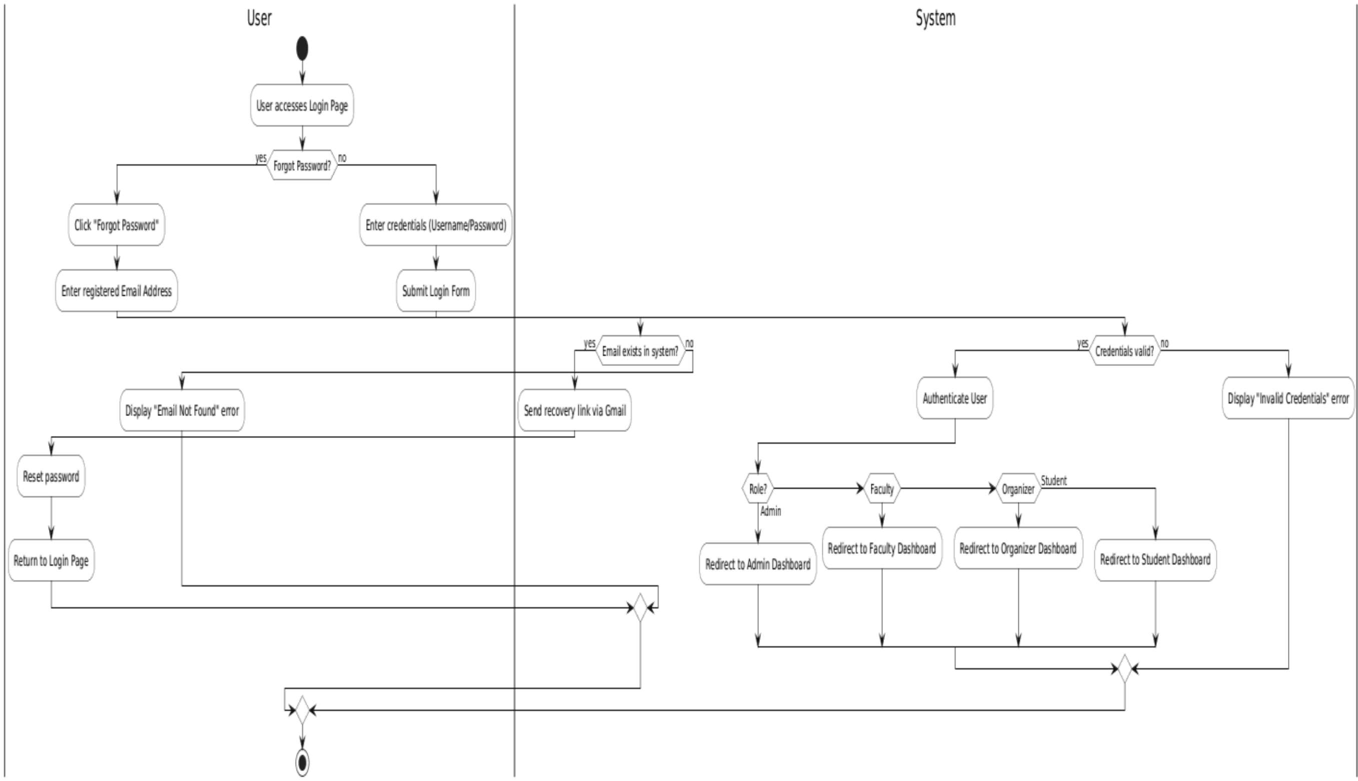


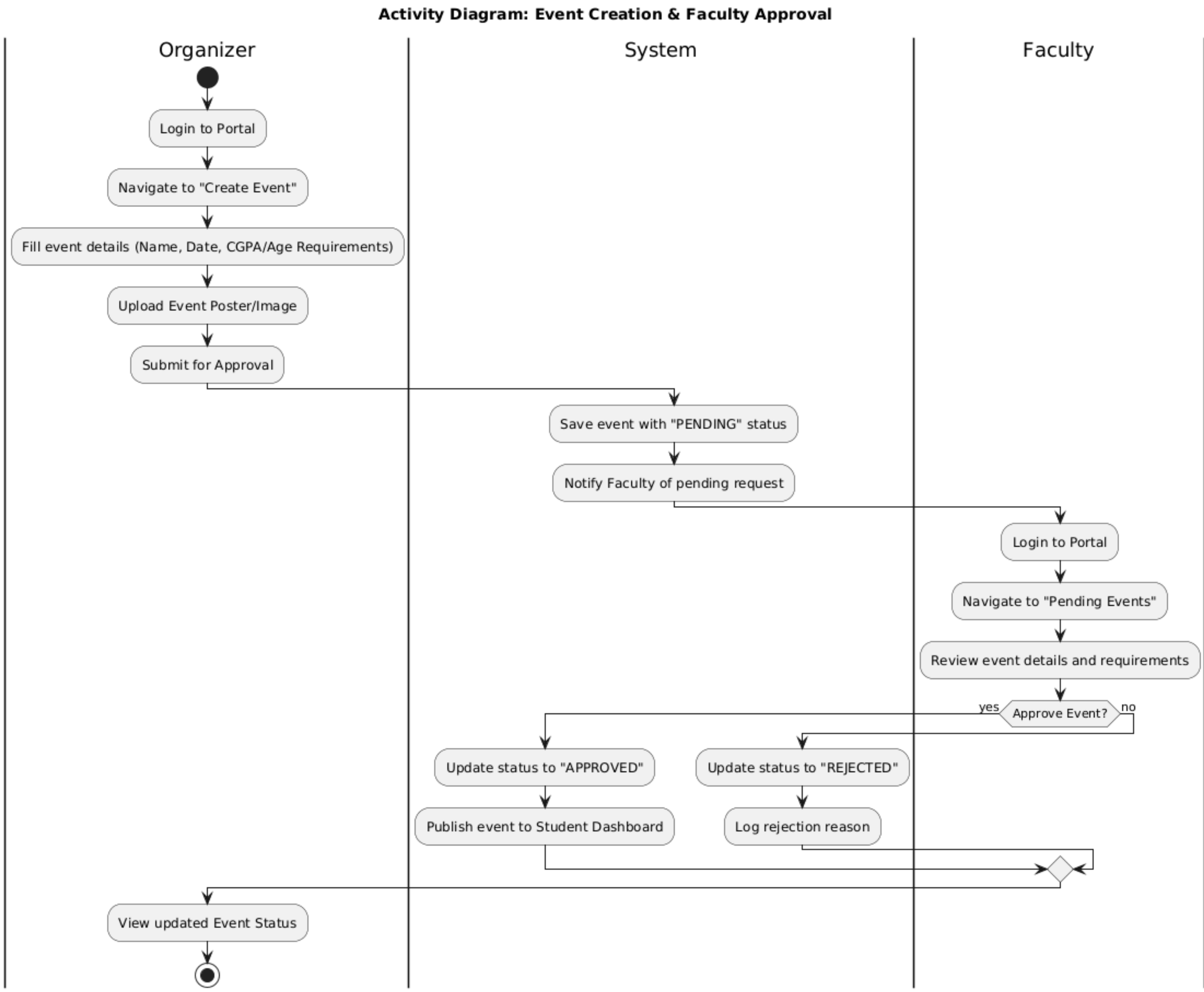
State Diagram: Payment Lifecycle



Activity Diagrams:

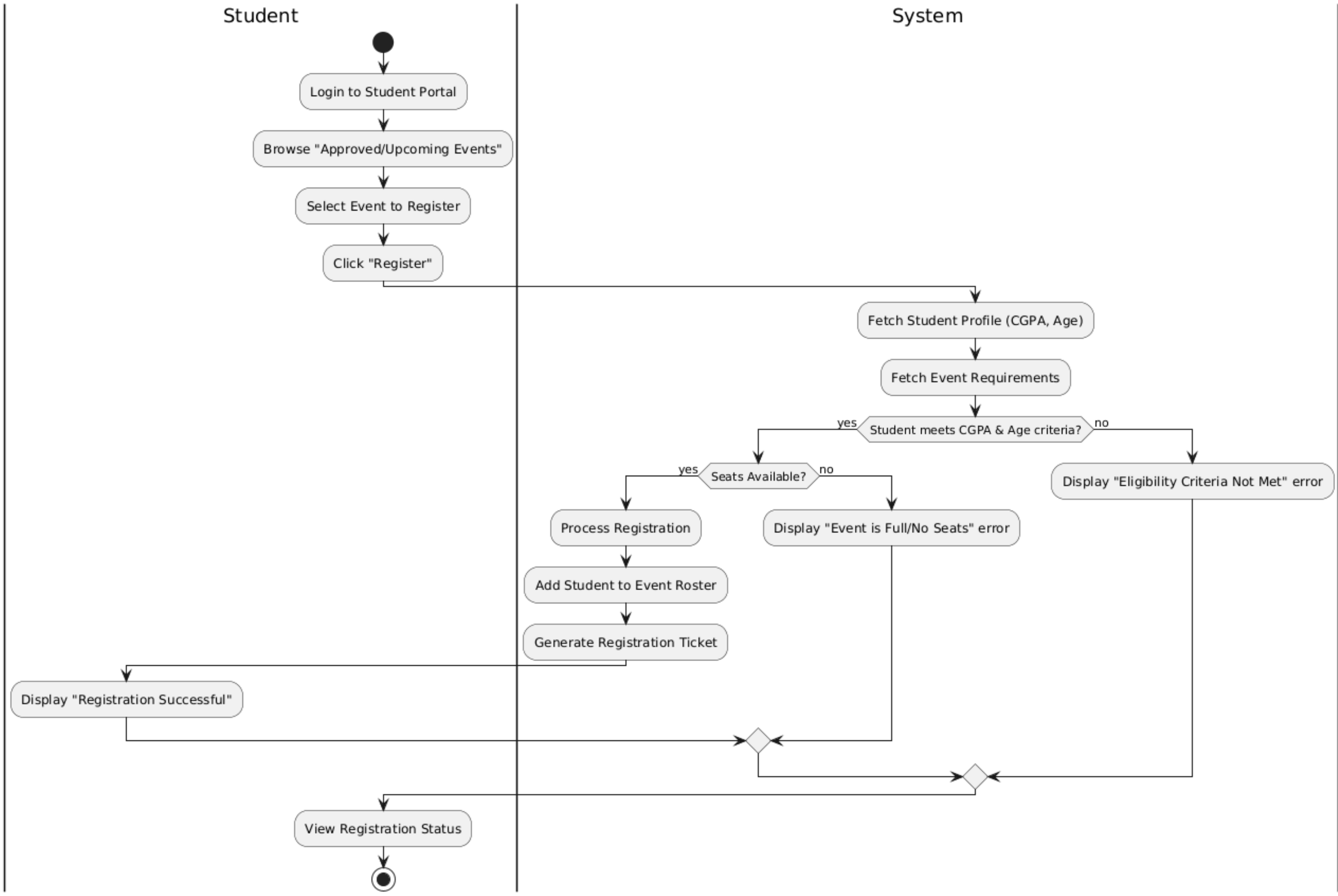
Activity Diagram: User Authentication & Recovery



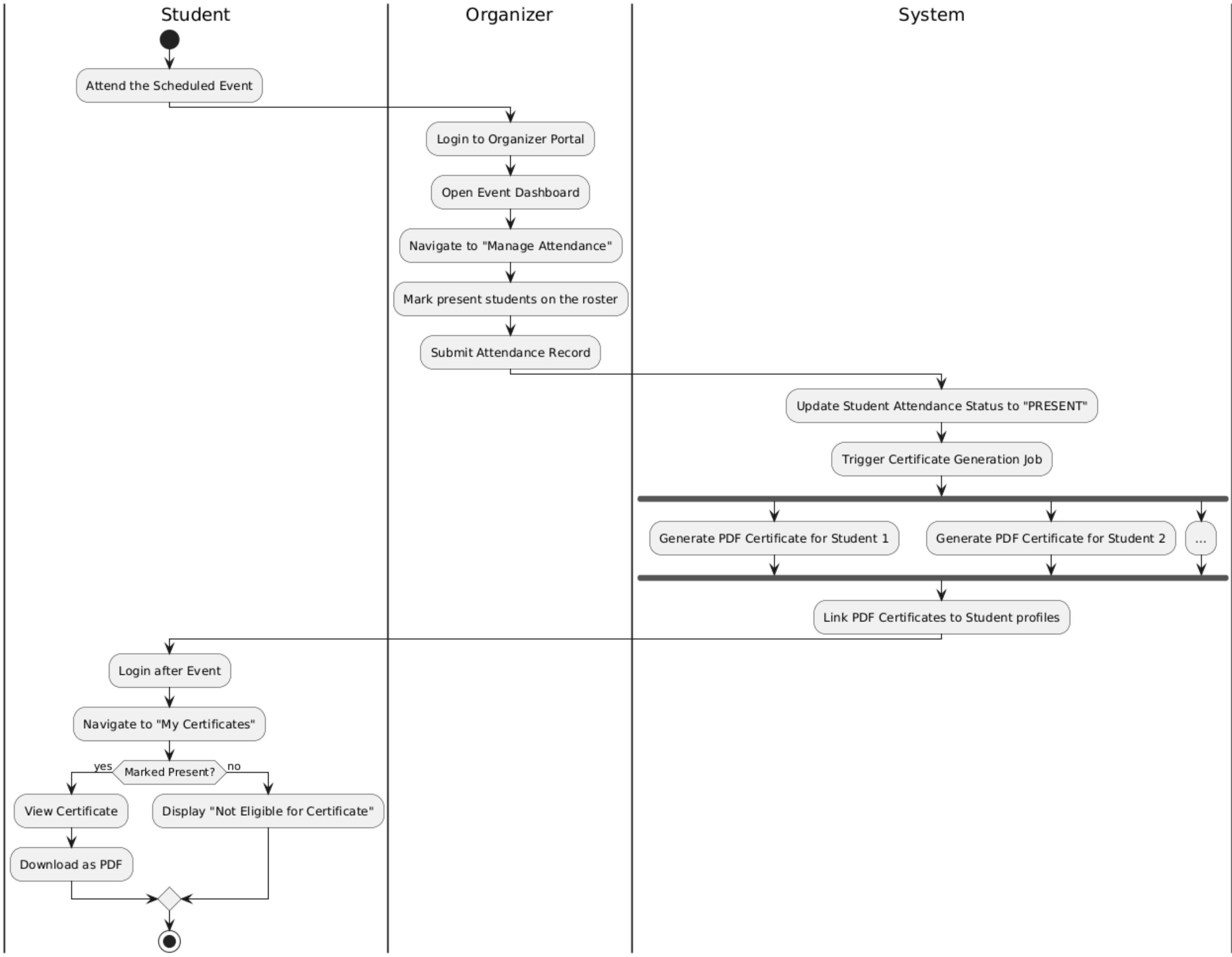


Activity Diagram: Event Creation & Faculty Approval
Organizer
System
Faculty
Login to Portal
Navigate to "Create Event"
Fill event details (Name, Date, CGPA/Age Requirements)
Upload Event Poster/Image
Submit for Approval
Save event with "PENDING" status
Notify Faculty of pending request
Login to Portal
Navigate to "Pending Events"
Review event details and requirements
Approve Event?
yes
no
Update status to "APPROVED"
Publish event to Student Dashboard
Update status to "REJECTED"
Log rejection reason
View updated Event Status

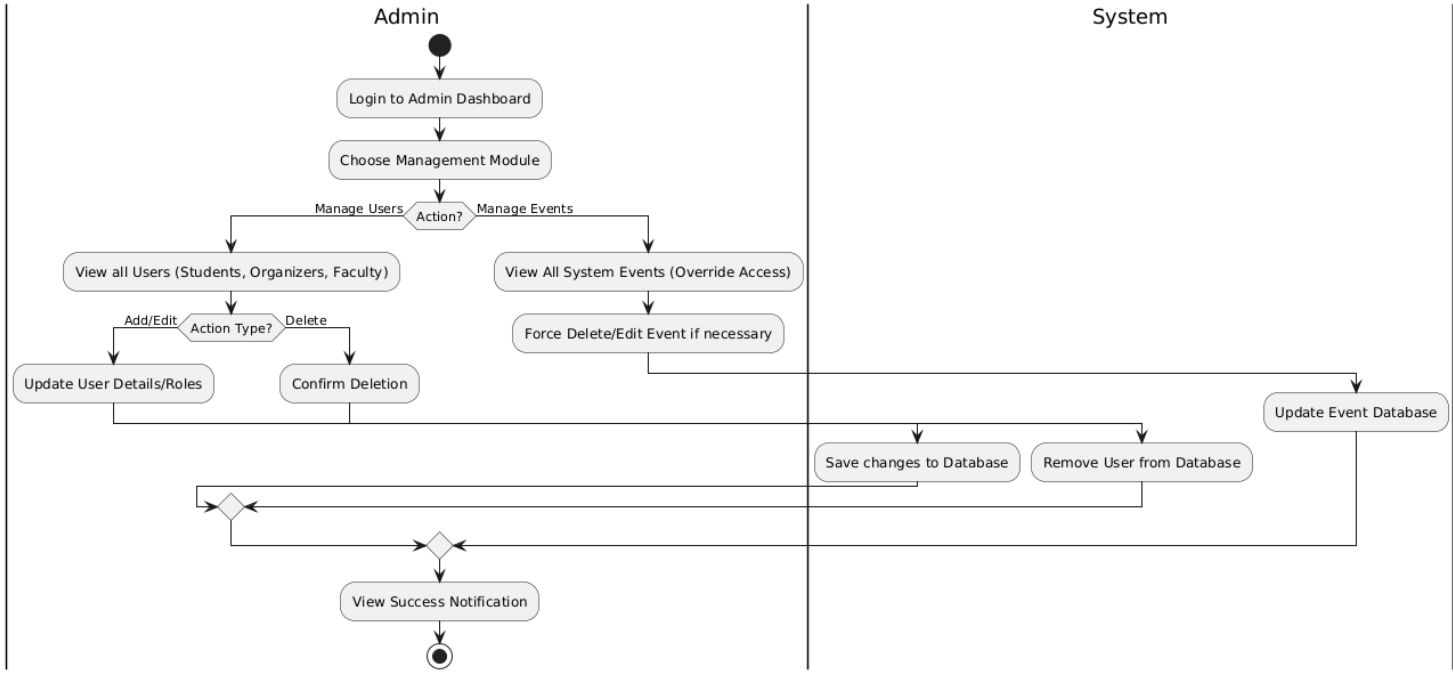
Activity Diagram: Student Event Registration



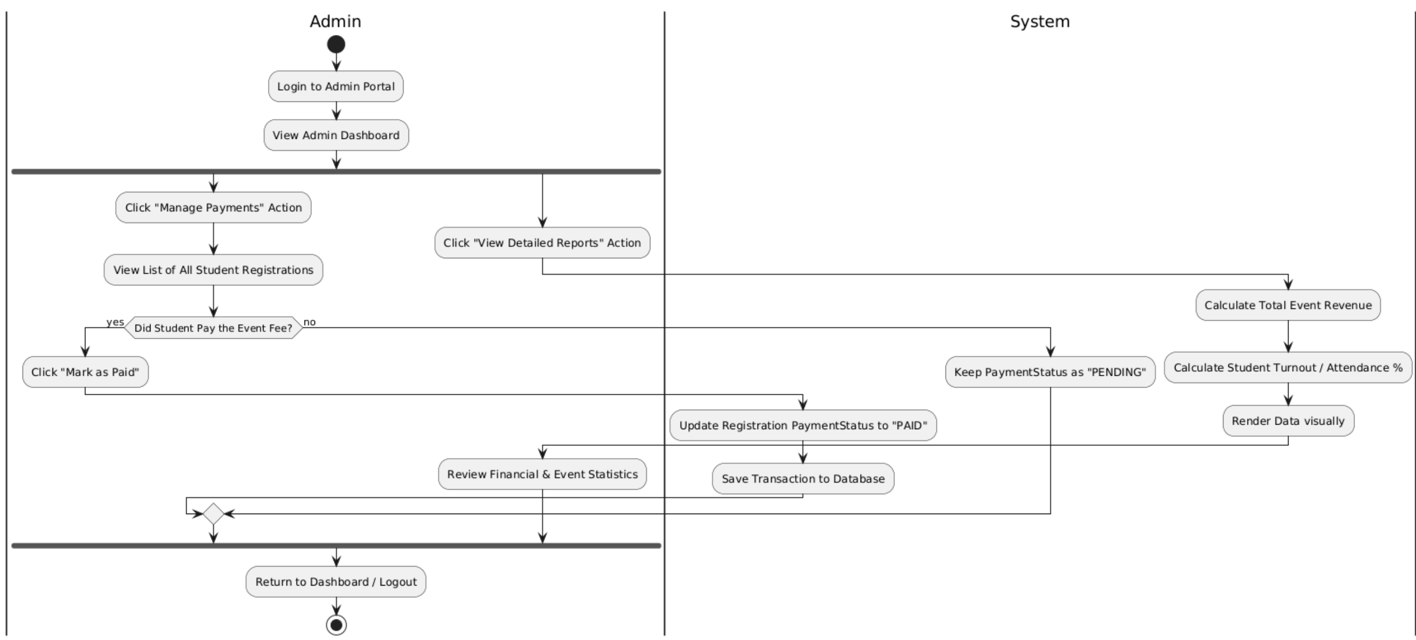
Activity Diagram: Attendance & Certificate Generation



Activity Diagram: Admin User Management



Activity Diagram: Admin Payments & Reports Workflow



Design Principles, and Design Patterns

MVC Architecture used? Yes

- ❖ The project is built upon the **Model-View-Controller (MVC)** architectural pattern using the Spring Boot framework. This pattern divides the application into three interconnected components, ensuring a clear separation of internal representations of information from the ways that information is presented to, and accepted from, the user.

- ❖ **Model (Data & Logic layer):** Represents the data and the core business logic of the application. In this project, the Model is implemented using JPA Entity classes (such as User, Event, and Registration). These classes map directly to the backend database tables. The Model also encompasses the Service layer (UserService, RegistrationService), which enforces business rules like checking student eligibility (CGPA and age) before allowing database updates.

- ❖ **View (Presentation layer):** Responsible for the user interface and displaying the data provided by the model. The project utilizes Thymeleaf HTML templates (e.g., layout.html, dashboard.html, register.html) as the View layer. It dynamically renders the data to the browser, displaying tailored dashboards based on the logged-in user's role (Admin, Faculty, Organizer, or Student).

- ❖ **Controller (Routing & Request layer):** Acts as an interface between the Model and the View. The Controllers (such as StudentController or HomeController) intercept incoming HTTP requests from the user's browser, process user inputs, invoke the necessary business logic from the Service layer, and return the appropriate Thymeleaf View populated with the required data object.

Design Principles:

- ❖ **Single Responsibility Principle (SRP):** The project architecture strictly separates concerns into distinct layers. \`Controllers\` (e.g., \`StudentController\`) handle only HTTP requests and routing, \`Services\` (e.g., \`RegistrationService\`) encapsulate core business logic, and \`Repositories\` manage database interactions. This ensures each class has only one reason to change.

- ❖ **Dependency Inversion Principle (DIP):** Through Spring Boot’s Inversion of Control (IoC) and Dependency Injection, high-level modules like Controllers do not depend on low-level concrete implementations. Instead, dependencies are injected via constructors or \`@Autowired\`, making the system loosely coupled and easier to test.

- **Don't Repeat Yourself (DRY):** Common, reusable logic—such as user authentication checks, role-based authorization, and student eligibility filtering (CGPA and age)—is centralized within dedicated service methods rather than being duplicated across multiple controller endpoints.

Design Patterns:

- **Model-View-Controller (MVC) Pattern:** This is the core architectural pattern of the application. The **Model** represents the data structures (Entities like `User`, `Event`, `Registration`), the **View** defines the frontend presentation (Thymeleaf HTML templates like `attendance.html`), and the **Controller** manages the data flow between them.

- **Singleton Pattern:** Managed implicitly by the Spring framework. Service classes (like `UserService`) and DAO Repositories are registered as Spring Beans with a default singleton scope. This ensures that only one instance of these classes exists system-wide, optimizing memory consumption and standardizing state.

- **Builder Pattern:** Utilized extensively for complex object creation. It is implemented specifically in `CertificateBuilder.java` to step-by-step construct and generate dynamic PDF certificates. It is also used across entity classes (via Lombok's `@Builder` annotation) to instantiate records cleanly without cumbersome constructors.

- **Repository / DAO Pattern:** Implemented using Spring Data JPA (`JpaRepository`). This pattern abstracts the underlying database interactions, allowing the application to perform CRUD (Create, Read, Update, Delete) operations using domain objects rather than raw, hard-coded SQL queries.

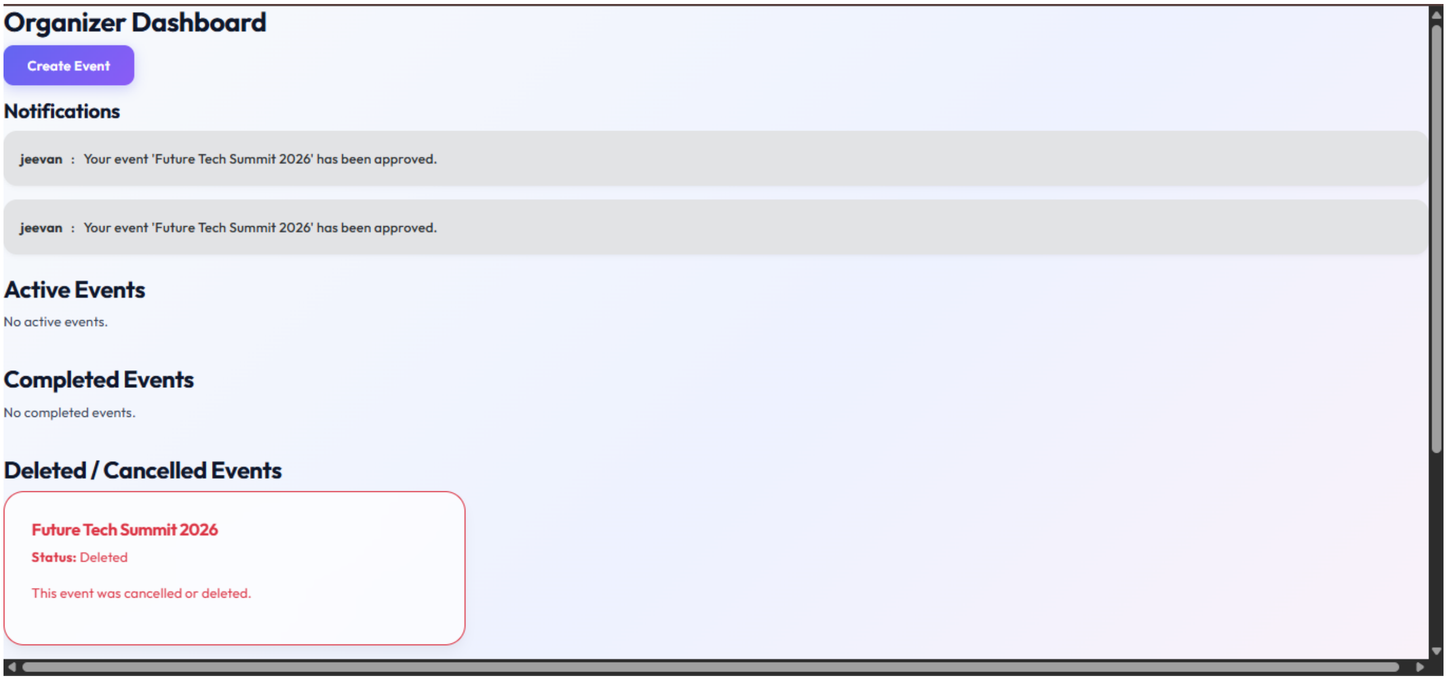
Github link to the Codebase:

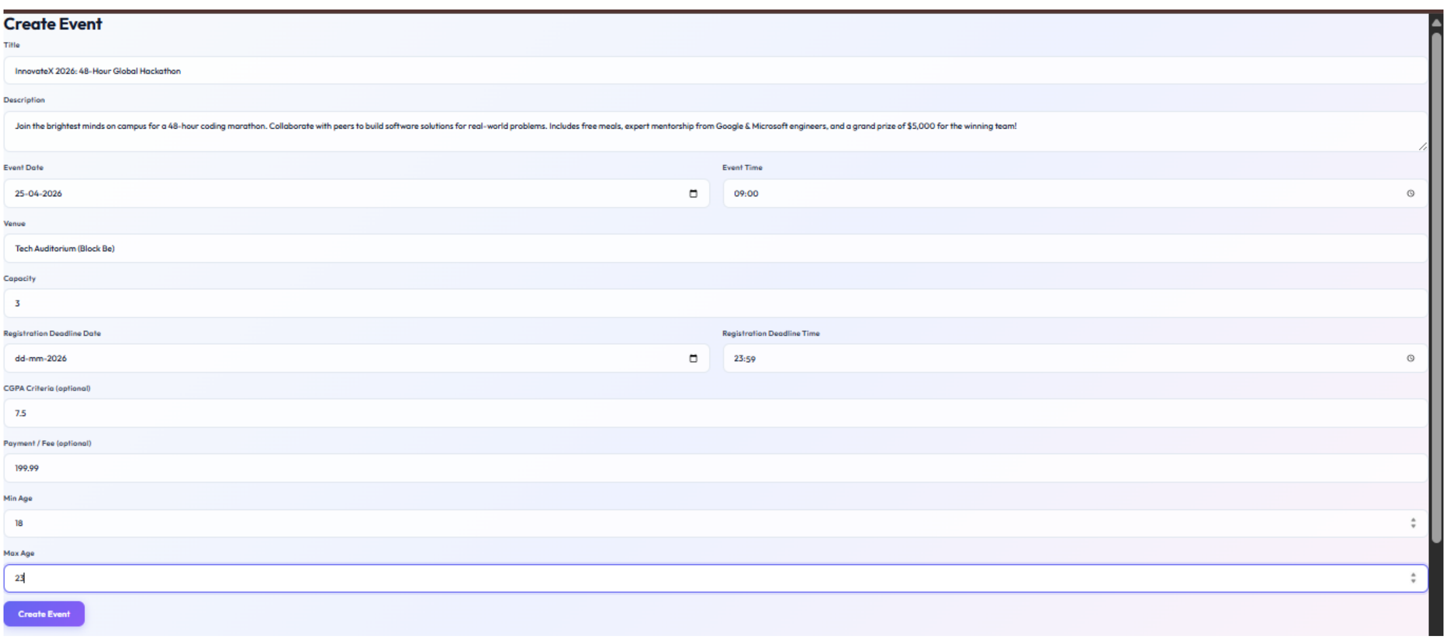
https://github.com/Keshavvs123/smart-campus-event-management-system

Screenshots

UI:

Organizer:






Organizer Dashboard
Create Event
Notifications
jeevan : Your event 'Future Tech Summit 2026' has been approved.
jeevan : Your event 'Future Tech Summit 2026' has been approved.
Active Events
InnovateX 2026: 48-Hour Global Hackathon
Join the brightest minds on campus for a 48-hour coding marathon. Collaborate with peers to build software solutions for real-world problems. Includes free meals, expert mentorship from Google & Microsoft engineers, and a grand prize of $5,000 for the winning team!
Status: Pending Faculty Approval
Edit
Delete
Registrations
Attendance
Completed Events
No completed events.
Deleted / Cancelled Events
Future Tech Summit 2026
Status: Deleted
This event was cancelled or deleted.
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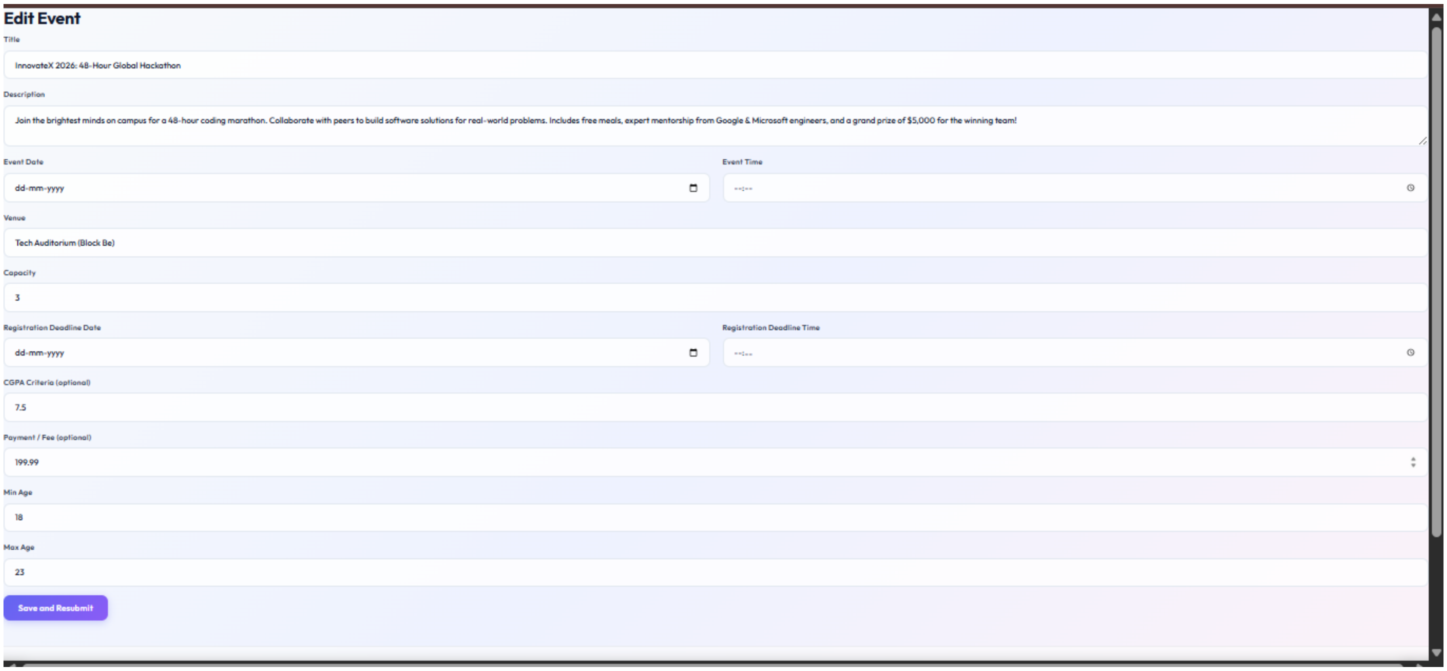
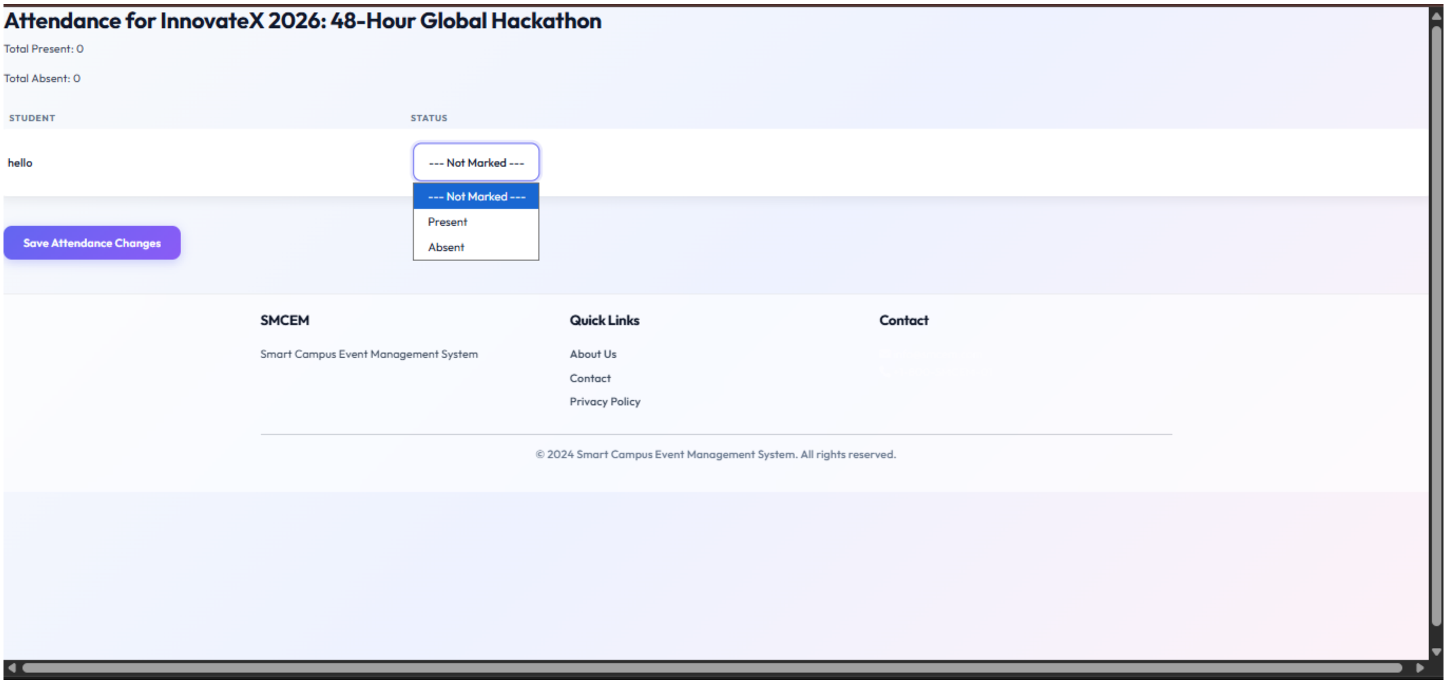

Edit Event
Title
InnovateX 2026: 48-Hour Global Hackathon
Description
Join the brightest minds on campus for a 48-hour coding marathon. Collaborate with peers to build software solutions for real-world problems. Includes free meals, expert mentorship from Google & Microsoft engineers, and a grand prize of $5,000 for the winning team!
Event Date
dd-mm-yyyy
Event Time
--:--
Venue
Tech Auditorium (Block 8e)
Capacity
3
Registration Deadline Date
dd-mm-yyyy
Registration Deadline Time
--:--
CGPA Criteria (optional)
7.5
Payment / Fee (optional)
199.99
Min Age
18
Max Age
23
Save and Resubmit


Attendance for InnovateX 2026: 48-Hour Global Hackathon
Total Present: 0
Total Absent: 0
STUDENT
STATUS
hello
--- Not Marked ---
--- Not Marked ---
Present
Absent
Save Attendance Changes
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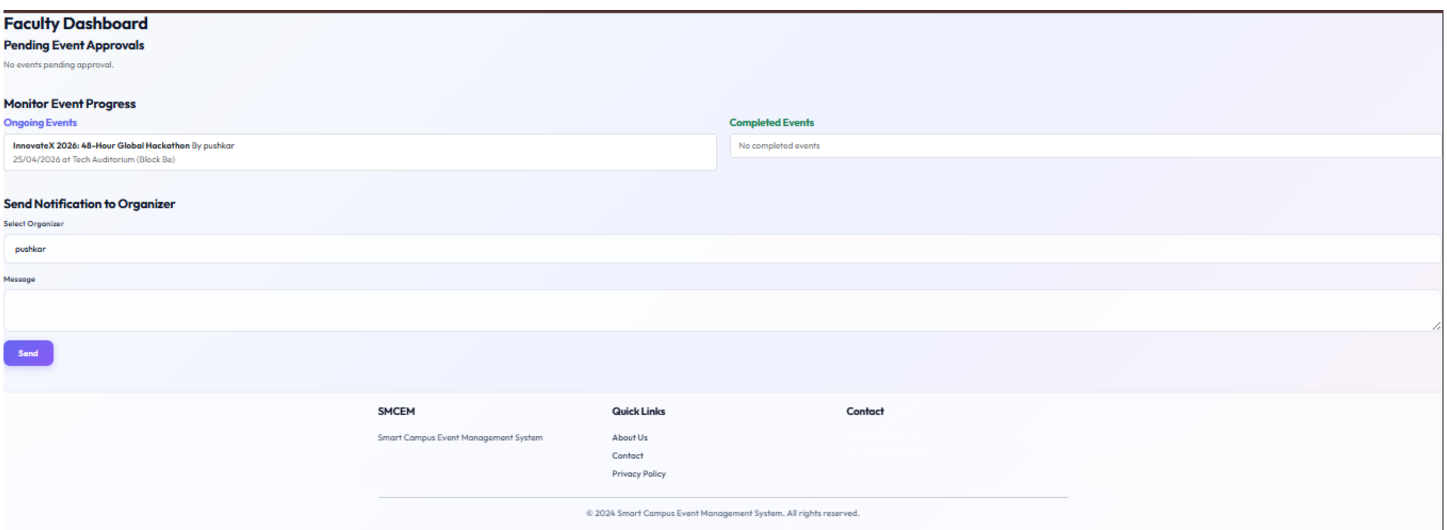
Faculty:


Faculty Dashboard
Pending Event Approvals
No events pending approval.
Monitor Event Progress
Ongoing Events
InnovateX 2026: 48-Hour Global Hackathon By pushkar
25/04/2026 at Tech Auditorium (Block Be)
Completed Events
No completed events
Send Notification to Organizer
Select Organizer
pushkar
Message
Send
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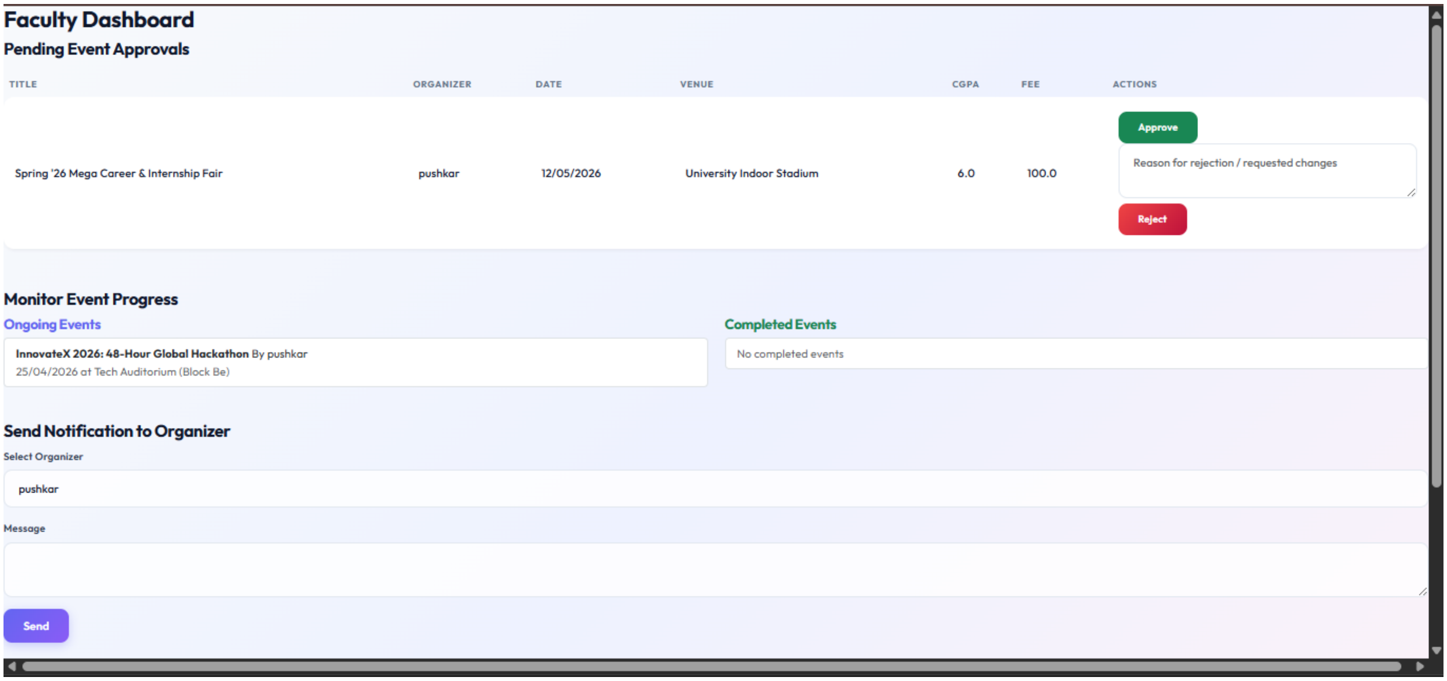
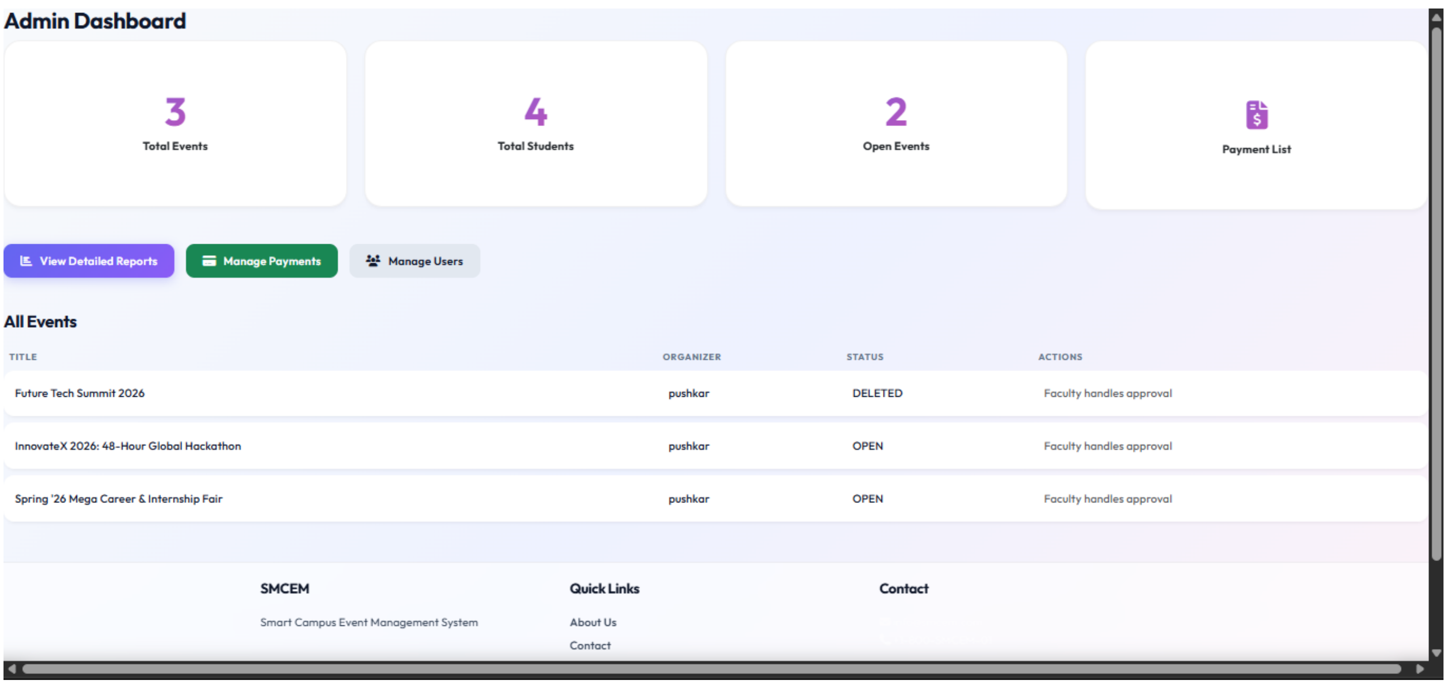

Faculty Dashboard
Pending Event Approvals
TITLE
ORGANIZER
DATE
VENUE
CGPA
FEE
ACTIONS
Spring '26 Mega Career & Internship Fair
pushkar
12/05/2026
University Indoor Stadium
6.0
100.0
Approve
Reason for rejection / requested changes
Reject
Monitor Event Progress
Ongoing Events
InnovateX 2026: 48-Hour Global Hackathon By pushkar
25/04/2026 at Tech Auditorium (Block Be)
Completed Events
No completed events
Send Notification to Organizer
Select Organizer
pushkar
Message
Send

Admin:


Admin Dashboard
3
Total Events
4
Total Students
2
Open Events
Payment List
View Detailed Reports
Manage Payments
Manage Users
All Events
TITLE
ORGANIZER
STATUS
ACTIONS
Future Tech Summit 2026
pushkar
DELETED
Faculty handles approval
InnovateX 2026: 48-Hour Global Hackathon
pushkar
OPEN
Faculty handles approval
Spring '26 Mega Career & Internship Fair
pushkar
OPEN
Faculty handles approval
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System Analytics & Reports

Back to Dashboard

Total Gross Revenue
$0,00

Number of Active Events
3

Event Participation Tracking

EVENT TITLE	CAPACITY	REGISTRATIONS	ACTUAL PRESENT	TURNOUT (%)	FEE SET	REVENUE ($)
Future Tech Summit 2026	4	2	0		$200.0	$0.0
InnovateX 2026: 48-Hour Global Hackathon	3	0	0		$199.99	$0.0
Spring '26 Mega Career & Internship Fair	2	0	0		$100.0	$0.0

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Manage Payments

Back to Dashboard

STUDENT NAME	EVENT	REQUIRED FEE	PAYMENT STATUS	ACTION
No payment records found.				

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Manage Users

Back to Dashboard

NAME	EMAIL	CURRENT ROLE	UPDATE ROLE		DANGER ZONE
keshav	keshav@gmail.com	STUDENT	Student	Update	Delete User
hello	hello@gmail.com	STUDENT	Student	Update	Delete User
pushkar	pushkar@gmail.com	ORGANIZER	Organizer	Update	Delete User
jeevan	jeevan@gmail.com	FACULTY	Faculty	Update	Delete User
guruvi	guruvi@gmail.com	ADMIN	Admin	Update	Delete User
hi	hi@gmail.com	STUDENT	Student	Update	Delete User
hey1	hey!@gmail.com	STUDENT	Student	Update	Delete User

Manage Payments

Back to Dashboard

STUDENT NAME	EVENT	REQUIRED FEE	PAYMENT STATUS	ACTION
hello	InnovateX 2026: 48-Hour Global Hackathon	199.99	PENDING	Mark as Paid
hey2	InnovateX 2026: 48-Hour Global Hackathon	199.99	PENDING	Mark as Paid
hey2	Spring '26 Mega Career & Internship Fair	100.0	PAID	Cleared

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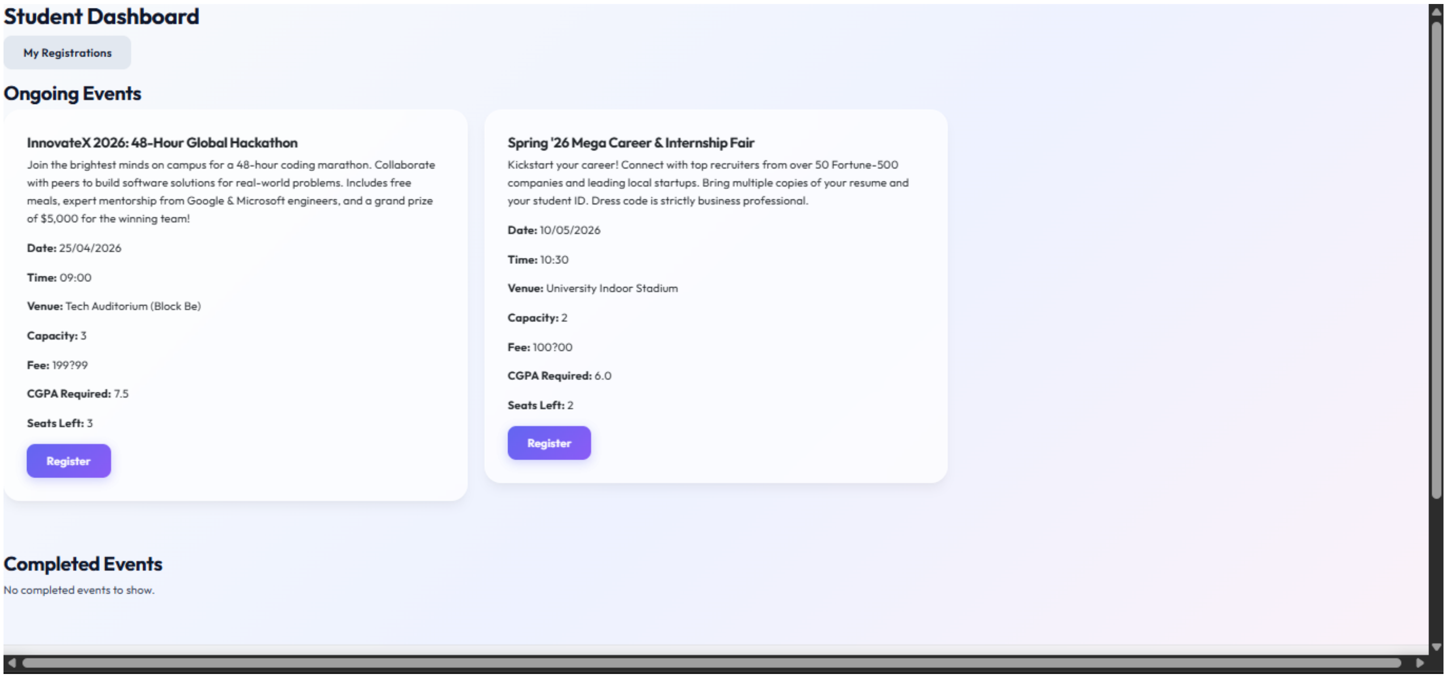
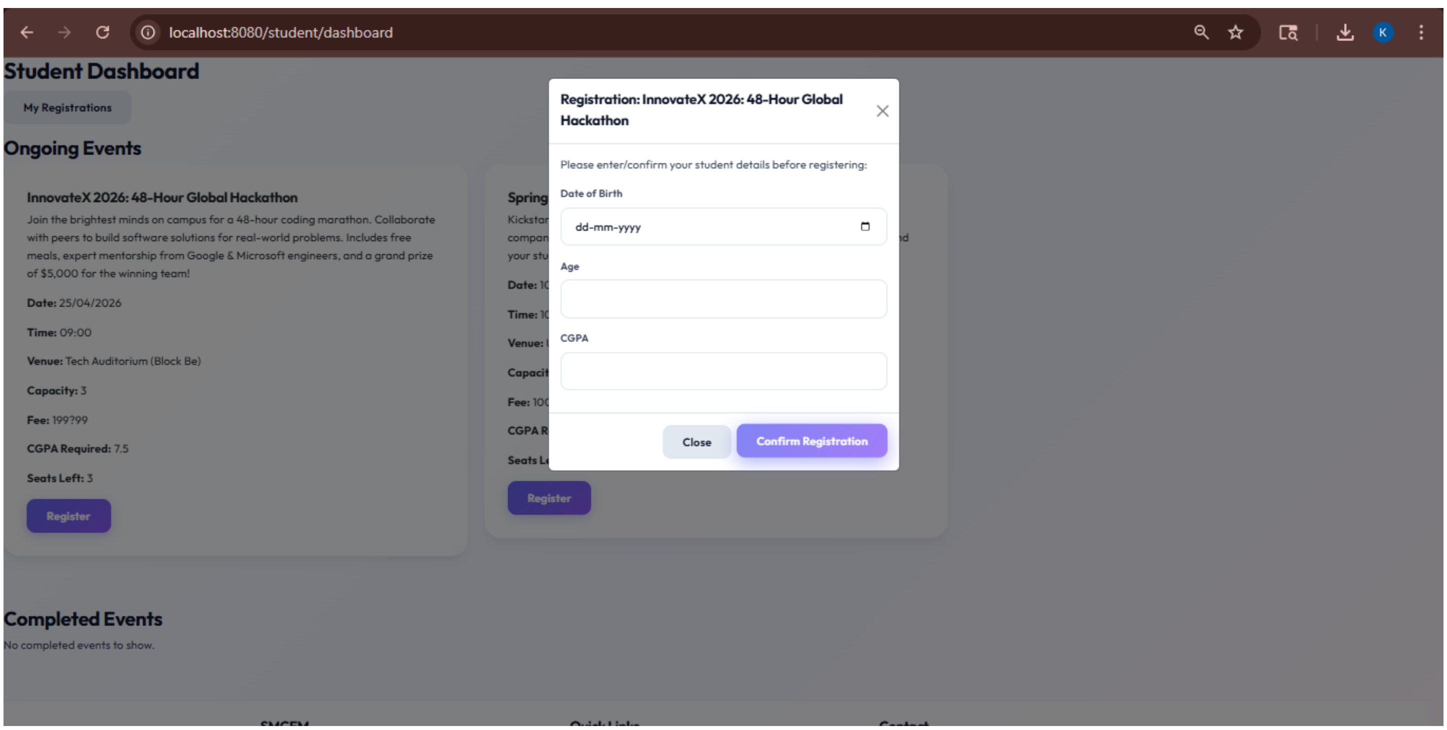
Activate Windows
Go to Settings to activate Windows.

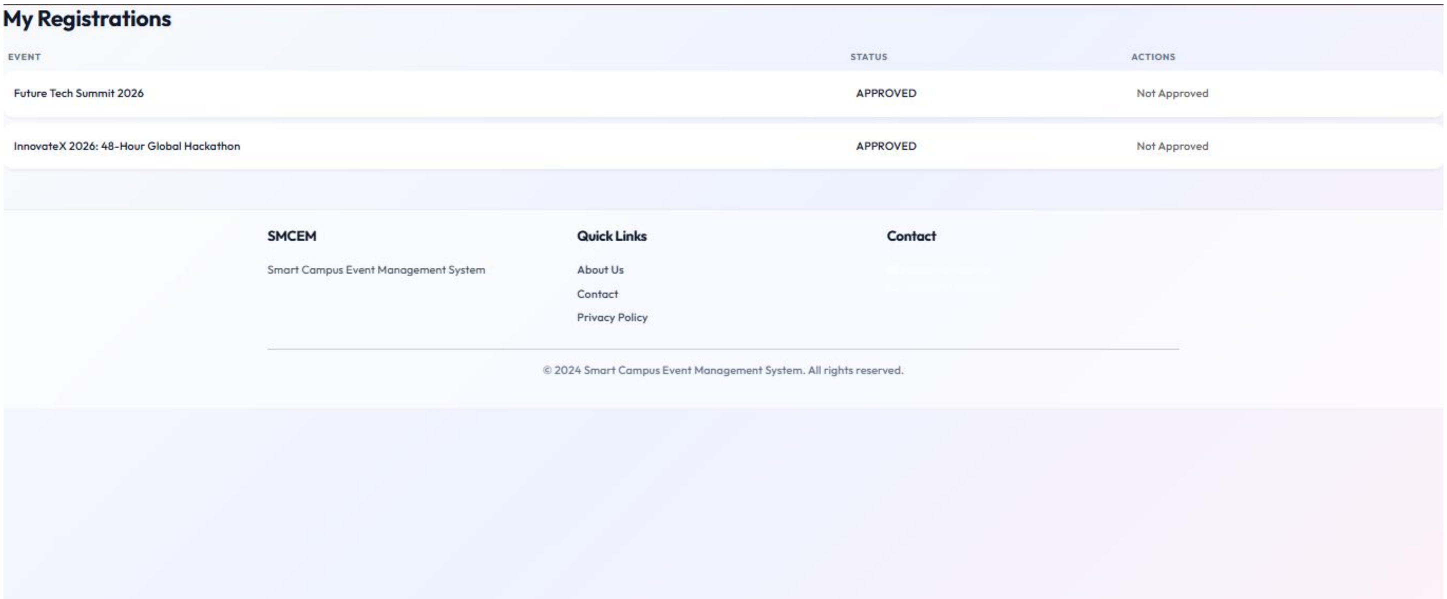
Student:



Student Dashboard
My Registrations
Ongoing Events
InnovateX 2026: 48-Hour Global Hackathon
Join the brightest minds on campus for a 48-hour coding marathon. Collaborate with peers to build software solutions for real-world problems. Includes free meals, expert mentorship from Google & Microsoft engineers, and a grand prize of $5,000 for the winning team!
Date: 25/04/2026
Time: 09:00
Venue: Tech Auditorium (Block Be)
Capacity: 3
Fee: 199?99
CGPA Required: 7.5
Seats Left: 3
Register
Spring '26 Mega Career & Internship Fair
Kickstart your career! Connect with top recruiters from over 50 Fortune-500 companies and leading local startups. Bring multiple copies of your resume and your student ID. Dress code is strictly business professional.
Date: 10/05/2026
Time: 10:30
Venue: University Indoor Stadium
Capacity: 2
Fee: 100?00
CGPA Required: 6.0
Seats Left: 2
Register
Completed Events
No completed events to show.



localhost:8080/student/dashboard
Student Dashboard
My Registrations
Ongoing Events
InnovateX 2026: 48-Hour Global Hackathon
Join the brightest minds on campus for a 48-hour coding marathon. Collaborate with peers to build software solutions for real-world problems. Includes free meals, expert mentorship from Google & Microsoft engineers, and a grand prize of $5,000 for the winning team!
Date: 25/04/2026
Time: 09:00
Venue: Tech Auditorium (Block Be)
Capacity: 3
Fee: 199?99
CGPA Required: 7.5
Seats Left: 3
Register
Registration: InnovateX 2026: 48-Hour Global Hackathon
Please enter/confirm your student details before registering:
Date of Birth
dd-mm-yyyy
Age
CGPA
Close
Confirm Registration
Register
Completed Events
No completed events to show.


My Registrations
EVENT
STATUS
ACTIONS
Future Tech Summit 2026
APPROVED
Not Approved
InnovateX 2026: 48-Hour Global Hackathon
APPROVED
Not Approved
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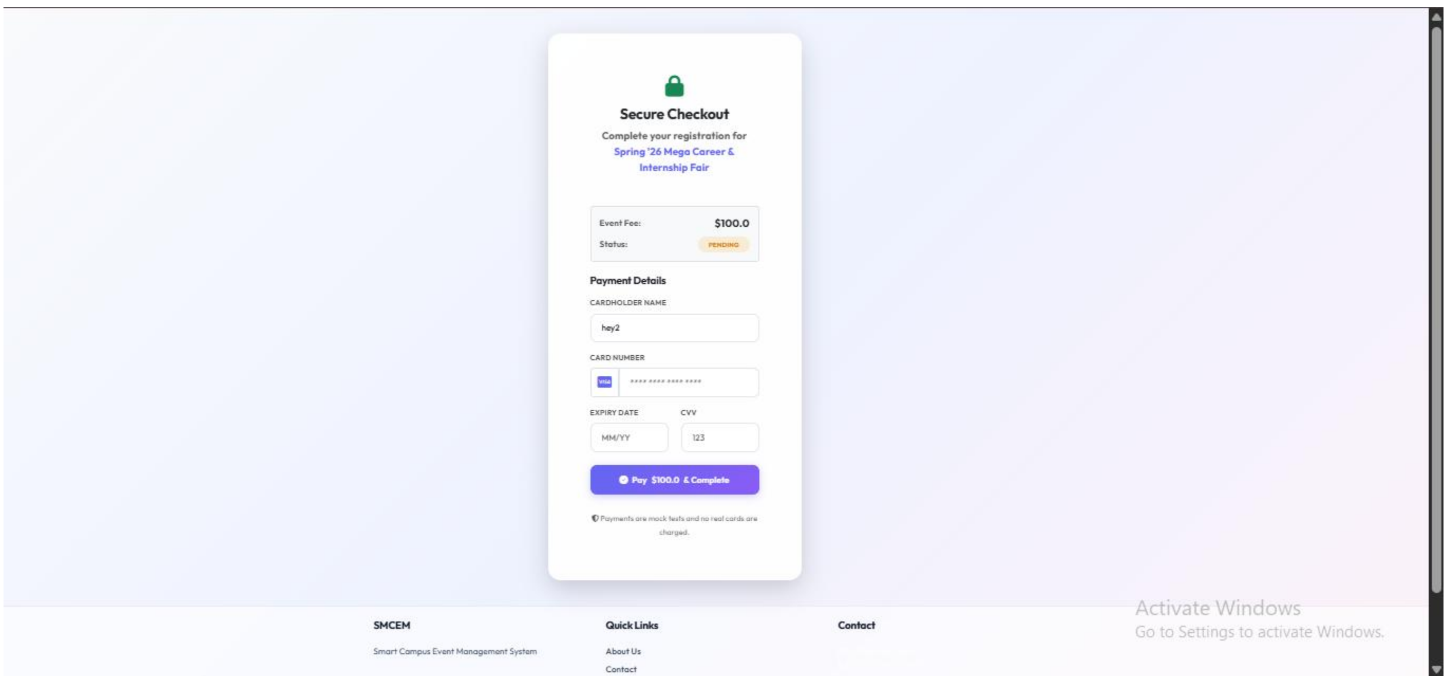
Secure Checkout
Complete your registration for
Spring '26 Mega Career & Internship Fair
Event Fee:
$100.0
Status:
PENDING
Payment Details
CARDHOLDER NAME
hey2
CARD NUMBER
VISA
**** **** **** ****
EXPIRY DATE
CVV
MM/YY
123
Pay $100.0 & Complete
Payments are mock tests and no real cards are charged.
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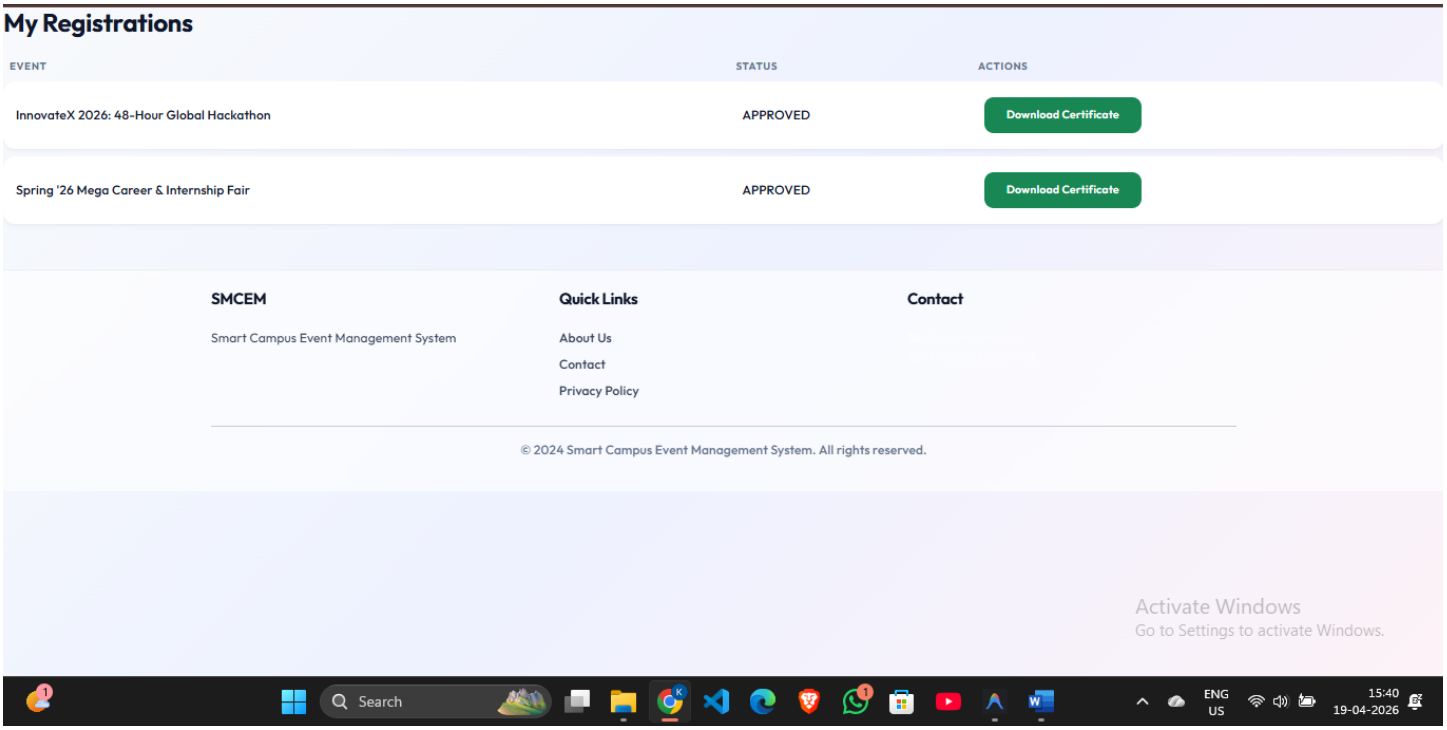
My Registrations
EVENT
STATUS
ACTIONS
InnovateX 2026: 48-Hour Global Hackathon
APPROVED
Download Certificate
Spring '26 Mega Career & Internship Fair
APPROVED
Download Certificate
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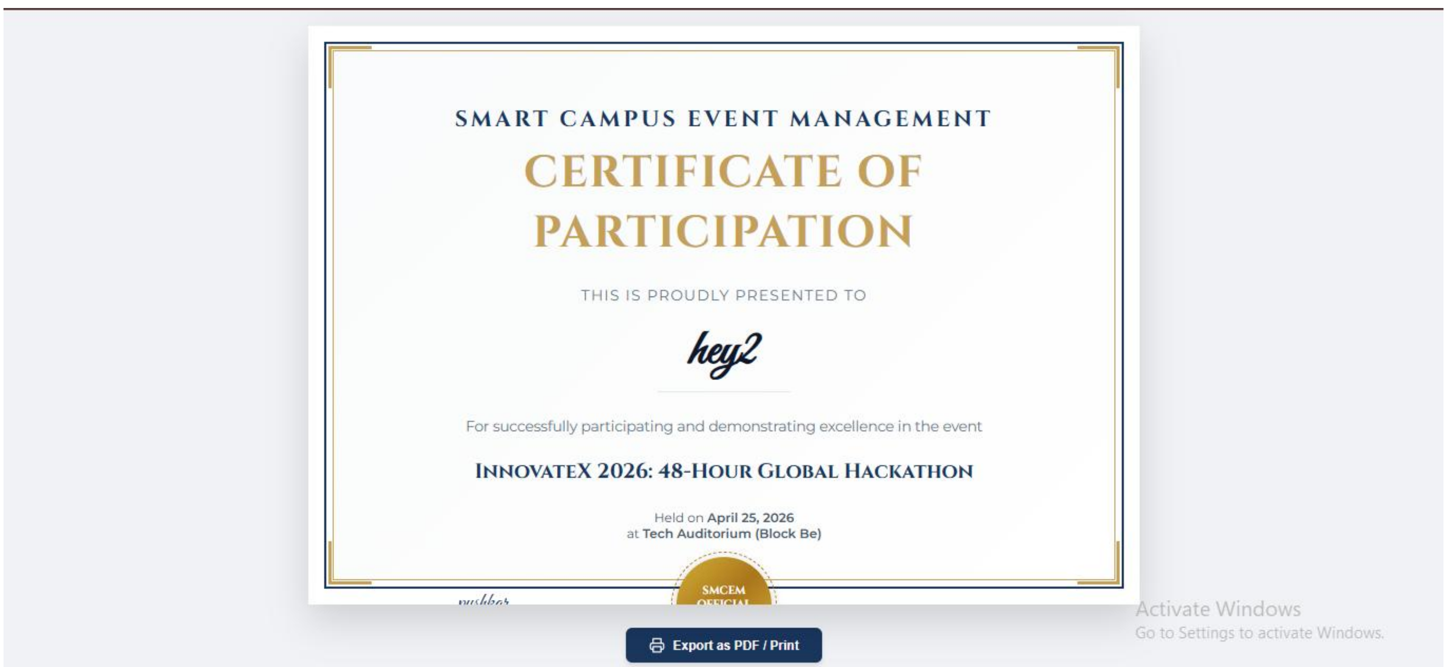
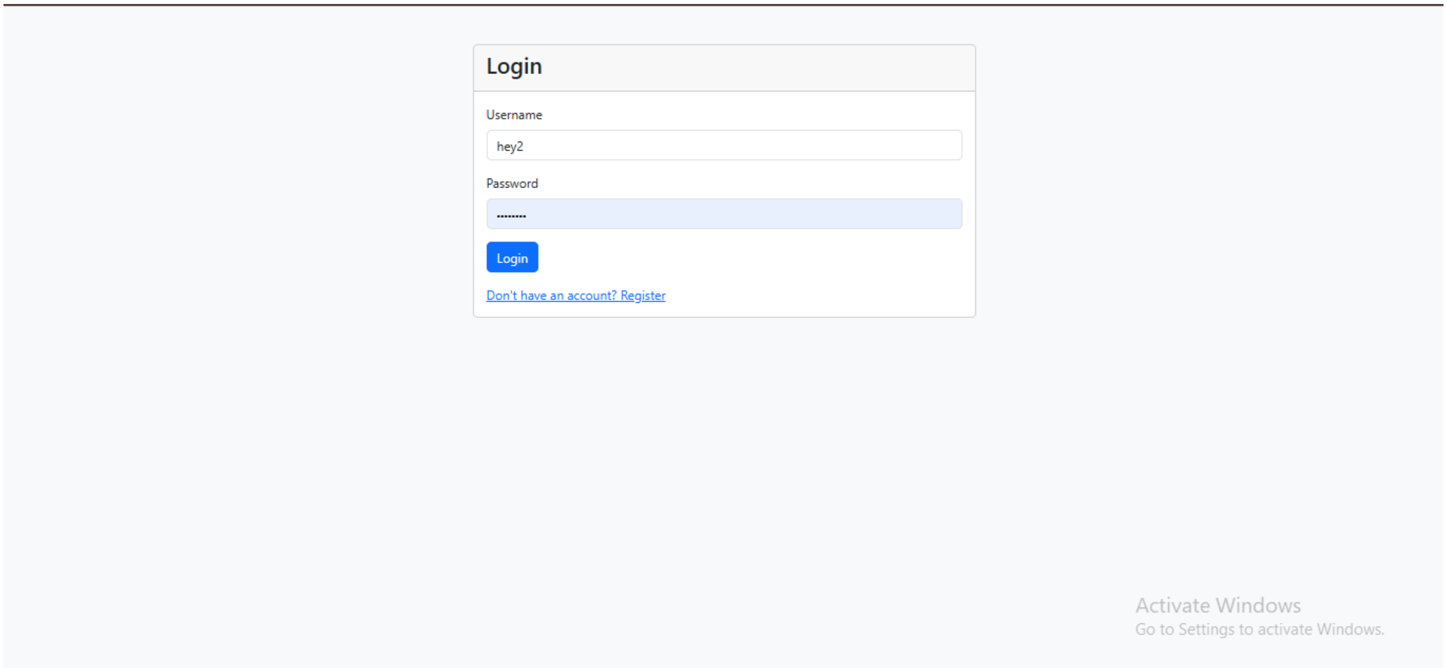
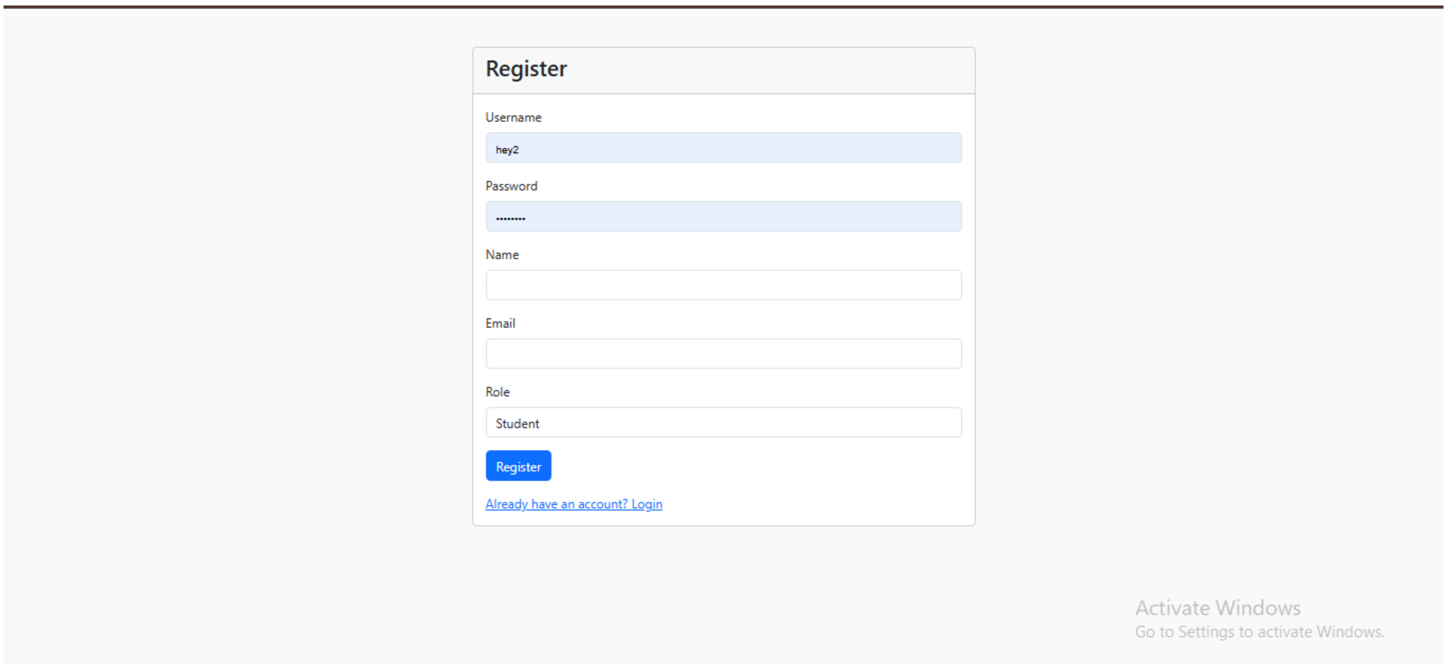
Activate Windows
Go to Settings to activate Windows.
Search
ENG
US
15:40
19-04-2026


SMART CAMPUS EVENT MANAGEMENT
CERTIFICATE OF PARTICIPATION
THIS IS PROUDLY PRESENTED TO
hey2
For successfully participating and demonstrating excellence in the event
INNOVATEX 2026: 48-HOUR GLOBAL HACKATHON
Held on April 25, 2026
at Tech Auditorium (Block Be)
SMCEM
Export as PDF / Print
Activate Windows
Go to Settings to activate Windows.

Login Page:



Registration Page:



Individual contributions of the team members:

Name	**Module worked on**
Keshav Vithal Sangonda	Student & Core Workflow Module: Implemented the Student dashboard, intelligent event filtering (Age/CGPA constraints), automated registration workflows, the secure mock payment gateway, and dynamic Certificate of Participation generation.
Pushkar P	Organizer & Event Management Module: Developed the Organizer dashboard, the complete event creation lifecycle (including UI for image uploading), seat capacity logic, and the real-time student attendance tracking system.
Jeevan A N	Faculty Review & Feedback Module: Designed the Faculty dashboard and the event approval ecosystem. Engineered the feedback loop allowing faculty to reject events and append required change messages directly to the Organizer.
Kasa Guruvi Reddy	Admin & Security Infrastructure: Built the core Spring Security authentication (including role-based access control and BCrypt hashing), the Admin overview dashboard with live statistics, global user management, and system reporting logic.